

Introduction

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UNIVERSIDAD
DE GRANADA



DaSCI

Instituto Andaluz de Investigación en
Data Science and Computational Intelligence

Index

1. What is Computer Vision?
2. Related Disciplines and Some Definitions
3. Why is Computer Vision Difficult?
4. Computer Vision Applications
5. A Bit of History
6. Some of the Main Current Limitations

What is computer vision?

- AI branch that deals with problems related to **visual perception**:
 - Science/Engineering dealing with the **automatic interpretation of images and videos** (i.e. to extract “meaning” from pixels)
 - Russell & Norvig’s AI book (Ch. 1) explicitly mentions 6 main AI branches:
 - Natural language processing
 - Knowledge representation
 - Automated reasoning
 - Machine learning
 - **Computer Vision (Ch. 27)**
 - Robotics
- } Necessary to pass the total Turing test, which requires interaction with objects and people in the real world

What is computer vision?

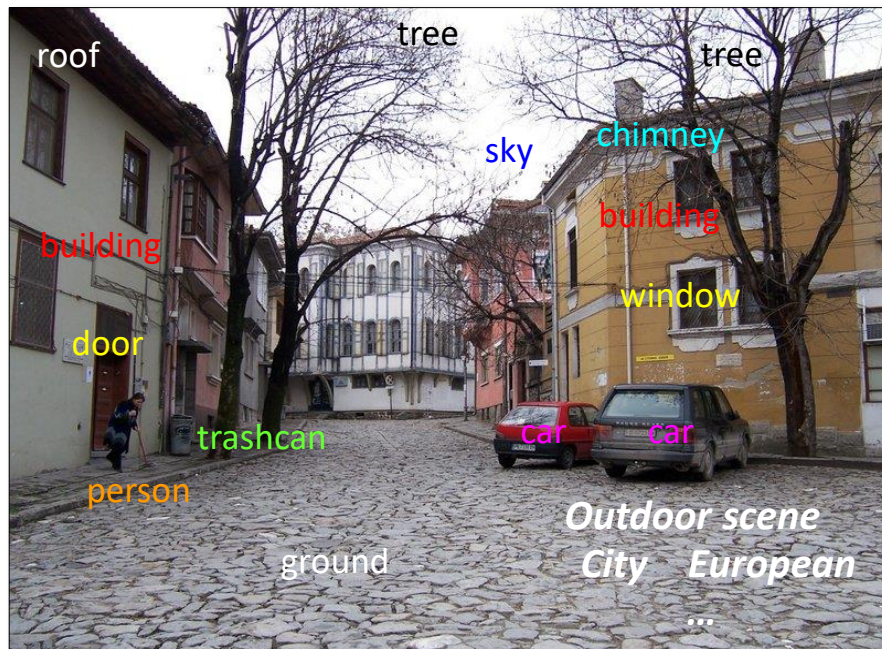
- “In computer vision, [we try] to **describe the world that we see** in one or more images and to **reconstruct its properties**, such as shape, illumination and color distributions.” Szeliski (2022)
- “Computer vision aims to duplicate the effect of human vision by electronically **perceiving and understanding an image**. [...] Giving computers the ability to see [...]” Sonka et al. (2015).
- “Computer vision aims at using **cameras** for analysing or **understanding scenes in the real world**.” Klette, R. *Concise computer vision*. Springer. 2014.

What is computer vision?

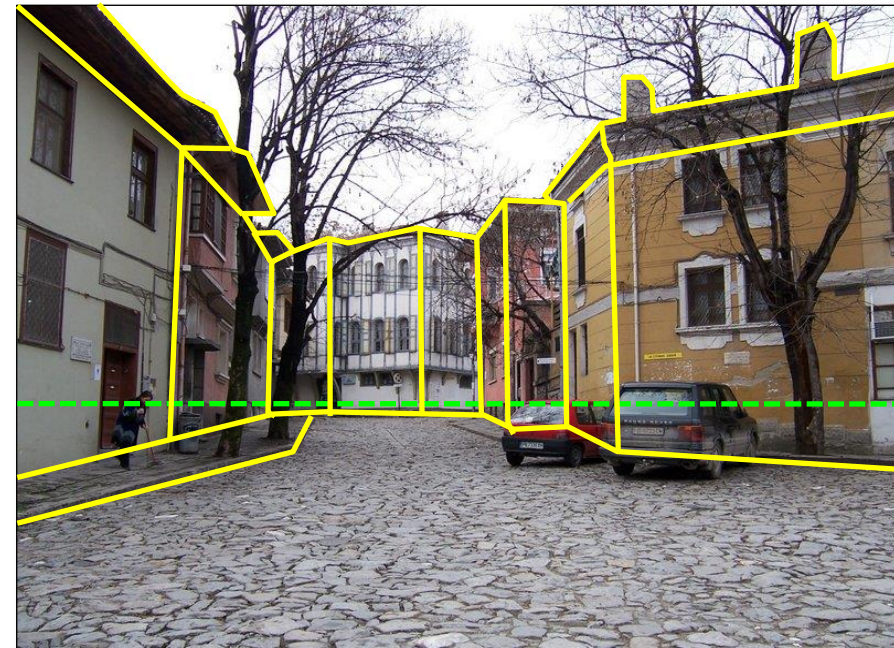
- “Computer vision [...] uses **statistical methods** to disentangle data using models constructed with the aid of **geometry, physics, and learning theory**. [...] Computer vision’s great trick is extracting **descriptions of the world from pictures** or sequences of pictures”. Forsyth, D. A., & Ponce, J. (2012).
- “How to **recover information** from the flood of data that comes **from eyes or cameras**” Russell & Norvig. *Artificial Intelligence: A Modern Approach*. Pearson. 2021.

What is computer vision?

- Automatic understanding of images and videos
 - Computing properties of the 3D world from visual data (*measurement*)
 - Algorithms and representations to allow a machine to recognize objects, people, scenes, and activities (*perception and interpretation*)

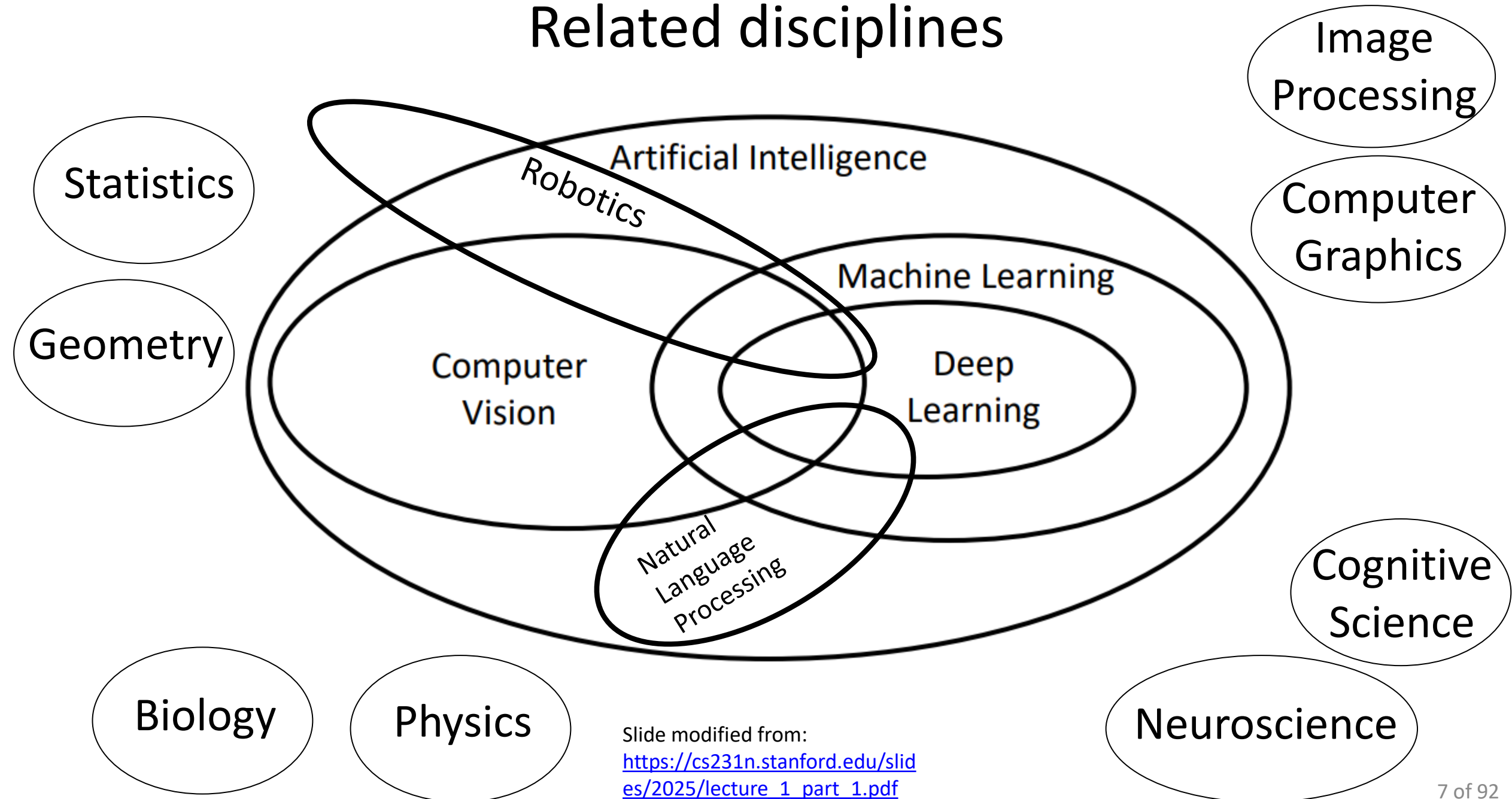


Semantic information



Geometric information

Related disciplines



Slide modified from:
https://cs231n.stanford.edu/slides/2025/lecture_1_part_1.pdf

Related disciplines

- Computer Vision vs
 - Computer Graphics (digital synthesis of visual content)



Image extracted from

https://upload.wikimedia.org/wikipedia/commons/5/5f/Utah_teapot_simple_2.png

- Image Processing (“low level” operations on images)



Image extracted from

<https://es.mathworks.com/help/images/ref/imgaussfilt.html>

Note: “machine vision” many times refers to the use of computer vision in industrial environments (for instance, automatic inspection, process control, etc.)

Vision and graphics

Images



Computer Vision



Model



Computer Graphics



Inverse problems: **analysis** and **synthesis**.

Slide credit: "Algorithms and applications in Computer Vision"
(Lihi Zelnik-Manor)

Low-, mid- and high-level processing

- Different authors employ different definitions and taxonomies
- Forsyth, D. A., & Ponce, J. (2012)
 - Early vision
 - Image filtering, texture analysis, stereopsis, structure from motion
 - Mid-level vision
 - It involves coming up with image representations that are simultaneously compact and expressive (image segmentation, tracking)
 - High-level vision
 - Image registration, image classification, object detection
- Sonka, M., Hlavac, V., & Boyle, R. (2015)
 - *Low-level image processing* methods usually use very little knowledge about the content of images.
 - Digital image processing: Image compression, pre-processing methods for noise filtering, edge extraction, and image sharpening
 - *High-level image understanding* tries to imitate human cognition. High-level data represent knowledge about the image content.

Higher level image
representation
↓

Higher level image
representation
↓

Low-, mid- and high-level processing

- Different authors employ different definitions and taxonomies
- Rafael C. González & Richard E. Woods (2008). Digital image processing. Pearson Prentice Hall. 3rd edition.
 - *Low-level processes* involve primitive operations (inputs and outputs are images)
 - image preprocessing to reduce noise, contrast enhancement, and image sharpening
 - *Mid-level processing* involves (inputs are images but outputs are attributes extracted from those images (e.g., edges, corners, or the identity of individual objects))
 - segmentation, objects description, and objects classification/recognition
 - *Higher-level processing* involves “making sense” of an ensemble of recognized objects and performing the cognitive functions normally associated with vision

Higher level image representation



Every image tells a story



- Goal of computer vision: perceive the “story” behind the picture
- Compute properties of the world
 - 3D shape
 - Names of people or objects
 - What happened?

Slide credit: Noah Snavely
(https://www.cs.cornell.edu/courses/cs5670/2023sp/lectures/lec00_intro_for_web.pdf)

Can computers match (or beat) human perception?



- Yes and no (mainly no)
 - computers can be better at “easy” things
 - humans are better at “hard” things
 - **What is considered “hard” keeps changing**
- But huge progress
 - Accelerated in the last 10-15 years due to **deep learning**

Slide credit: Noah Snaveley
(https://www.cs.cornell.edu/courses/cs5670/2023sp/lectures/lec00_intro_for_web.pdf)

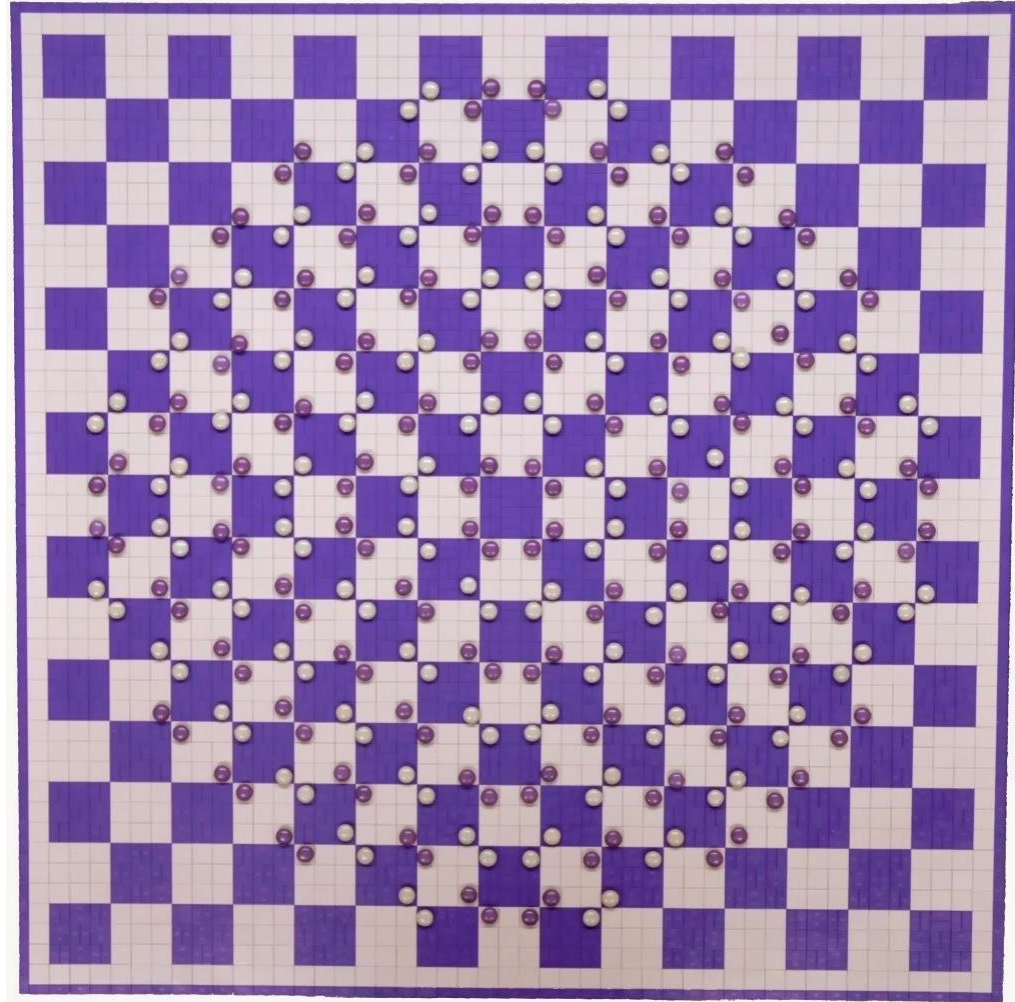
Can computers match (or beat) human perception?

– **What is considered
“hard” keeps changing**



“People who have not worked in the field often underestimate the difficulty of the problem. This misperception that vision should be easy dates back to the early days of artificial intelligence, when it was initially believed that the cognitive (logic proving and planning) parts of intelligence were intrinsically more difficult than the perceptual components.” Szeliski (2022)

Human perception has its shortcomings



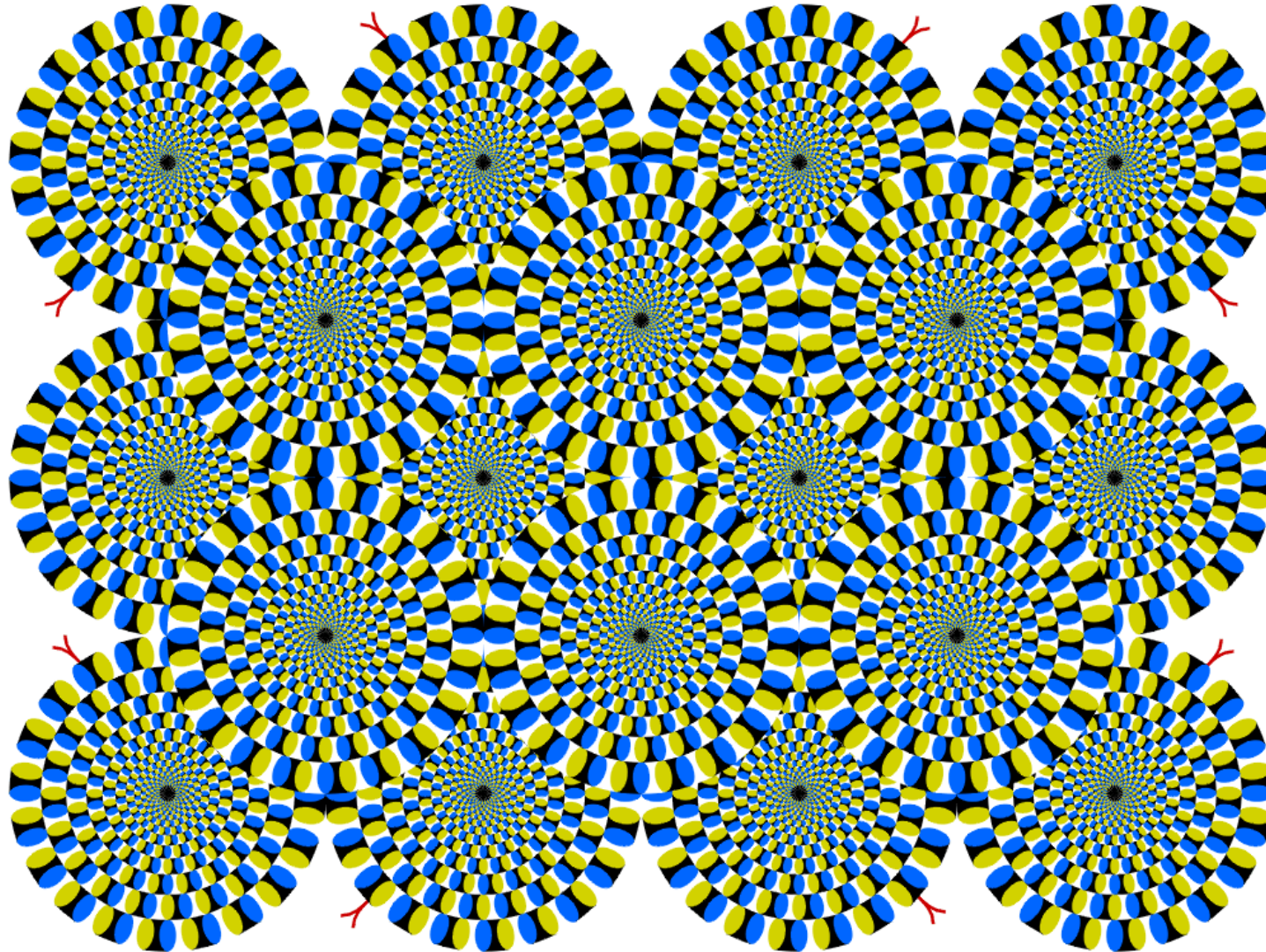
<https://twitter.com/pickover/status/1460275132958662657/>

Reality shatter. The physics, biology, and mathematics of human vision as it relates to geometrical patterns.

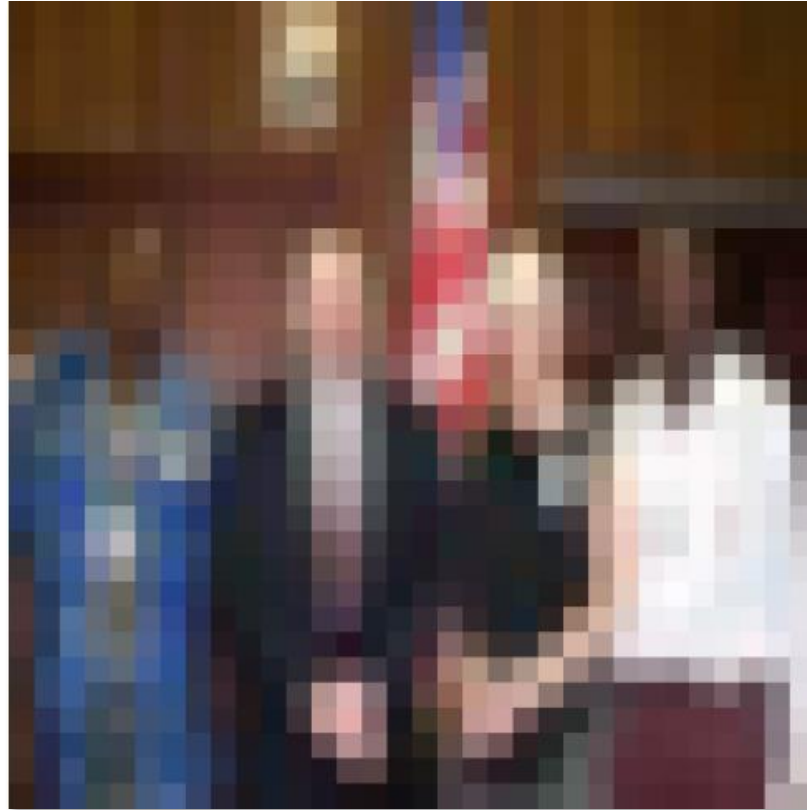
The lines on this checkerboard pattern are straight. (Optical illusion by Akiyoshi Kitaoka), Source: <https://bit.ly/3wNxuGQ>

Slide modified from Noah Snaveley
(https://www.cs.cornell.edu/courses/cs5670/2023sp/lectures/lec00_intro_for_web.pdf)

Human perception has its shortcomings



But humans can tell a lot about a scene from a little information...



Source: “80 million tiny images” by Torralba, et al.

Slide credit: Noah Snavely
(https://www.cs.cornell.edu/courses/cs5670/2023sp/lectures/lec00_intro_for_web.pdf)

Why is computer vision difficult?

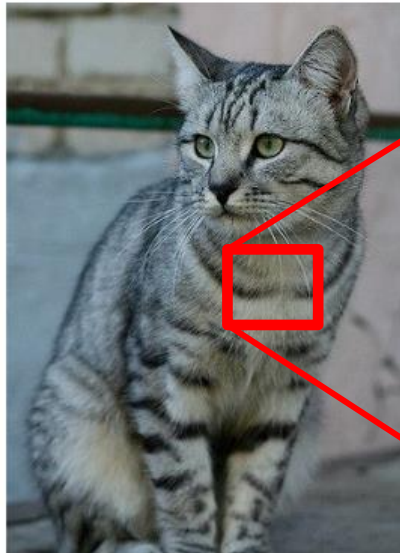
- We just take an example: Image Classification
 - Input: a single image
 - Output: Is there a cat in there? Yes/No?



This image by Nikita is
licensed under CC-BY 2.0

→ **CAT**

The problem: semantic gap



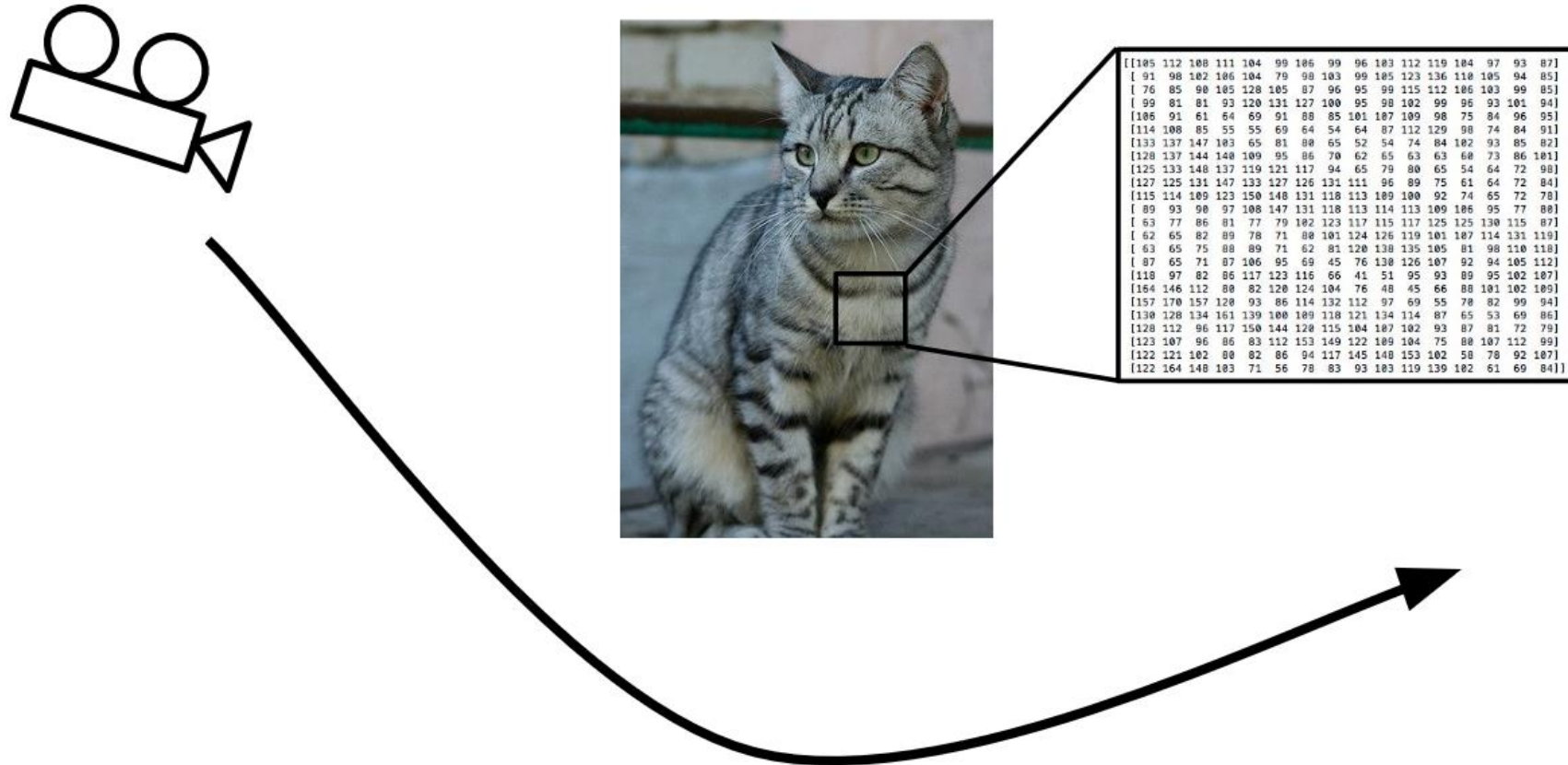
This image by Nikita is
licensed under CC-BY 2.0

We perceive an image
(photons processed by our brain).
Holistic view of a cat.

```
[[105 112 108 111 104 99 106 99 96 103 112 119 104 97 93 87]  
[ 91 98 102 106 104 79 98 103 99 105 123 136 110 105 94 85]  
[ 76 85 90 105 128 105 87 96 95 99 115 112 106 103 99 85]  
[ 99 81 81 93 120 131 127 100 95 98 102 99 96 93 101 94]  
[106 91 61 64 69 91 88 85 101 107 109 98 75 84 96 95]  
[114 108 85 55 55 69 64 54 64 87 112 129 98 74 84 91]  
[133 137 147 103 65 81 80 65 52 54 74 84 102 93 85 82]  
[128 137 144 140 109 95 86 70 62 65 63 63 60 73 86 101]  
[125 133 148 137 119 121 117 94 65 79 80 65 54 64 72 98]  
[127 125 131 147 133 127 126 131 111 96 89 75 61 64 72 84]  
[115 114 109 123 150 148 131 118 113 109 100 92 74 65 72 78]  
[ 89 93 90 97 108 147 131 118 113 114 113 109 106 95 77 80]  
[ 63 77 86 81 77 79 102 123 117 115 117 125 125 130 115 87]  
[ 62 65 82 89 78 71 80 101 124 126 119 101 107 114 131 119]  
[ 63 65 75 88 89 71 62 81 120 138 135 105 81 98 110 118]  
[ 87 65 71 87 106 95 69 45 76 130 126 107 92 94 105 112]  
[118 97 82 86 117 123 116 66 41 51 95 93 89 95 102 107]  
[164 146 112 80 82 120 124 104 76 48 45 66 88 101 102 109]  
[157 170 157 120 93 86 114 132 112 97 69 55 70 82 99 94]  
[130 128 134 161 139 100 109 118 121 134 114 87 65 53 69 86]  
[128 112 96 117 150 144 120 115 104 107 102 93 87 81 72 79]  
[123 107 96 86 83 112 153 149 122 109 104 75 80 107 112 99]  
[122 121 102 80 82 86 94 117 145 148 153 102 58 78 92 107]  
[122 164 148 103 71 56 78 83 93 103 119 139 102 61 69 84]]
```

Our computer just “sees” a set of numbers (a matrix).
An image is just a big grid of numbers between [0, 255]:
e.g. 800 x 600 x 3 (3 channels RGB)

Challenges: Viewpoint variation



All pixels change when the camera moves!

Challenges: Illumination



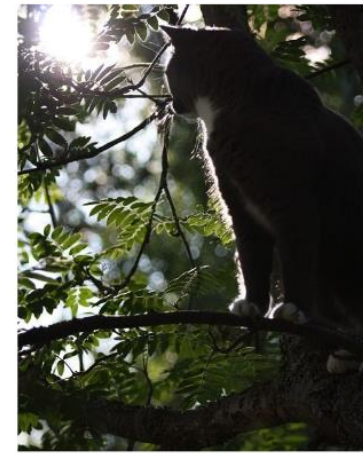
[This image is CC0 1.0 public domain](#)



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Challenges: Deformation



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This image by [sare bear](#) is licensed under [CC-BY 2.0](#)

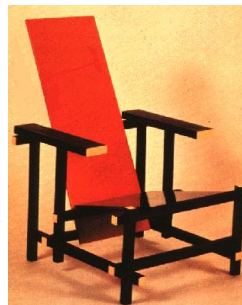


This image by [Tom Thai](#) is licensed under [CC-BY 2.0](#)

Challenges: Intraclass variation



This image is [CC0 1.0](#) public domain



slide credit: Fei-Fei, Fergus & Torralba

Challenges: Background Clutter



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[This image](#) is [CC0 1.0](#) public domain

Challenges: Occlusion



[This image](#) is [CC0 1.0](#) public domain



[This image](#) is [CC0 1.0](#) public domain



[This image](#) by [jonsson](#) is licensed under [CC-BY 2.0](#)

How on earth can a human know that there's a cat in there?



Challenges: Occlusion

ChatGPT 5.2 ▾



Descríbeme lo que ves en la siguiente imagen.

Veo un sofá de cuero color beige con varios cojines acolchados. En el asiento central, uno de los cojines está ligeramente levantado o desplazado, y por una abertura entre los cojines asoma una cola peluda con anillos oscuros y claros, que parece ser la cola de un mapache. A la derecha también se alcanza a ver un pequeño mechón de pelo blanco sobresaliendo. Da la impresión de que el animal está escondido dentro del sofá.



Challenges: occlusion, clutter



Challenges: Motion



How do we know that this isn't a headless bird? Why do we infer that there is movement?



Source: S. Lazebnik

Challenges: Inherent ambiguity and scale

- Many different 3D scenes could have given rise to a particular 2D picture



- We often must use **prior knowledge** about the **world's structure**

There are lots of visual cues we can use...

Depth cues: Linear perspective



Depth cues: Aerial perspective



Depth ordering cues



Shape cues: Texture gradient



Shape and lighting cues: Shading

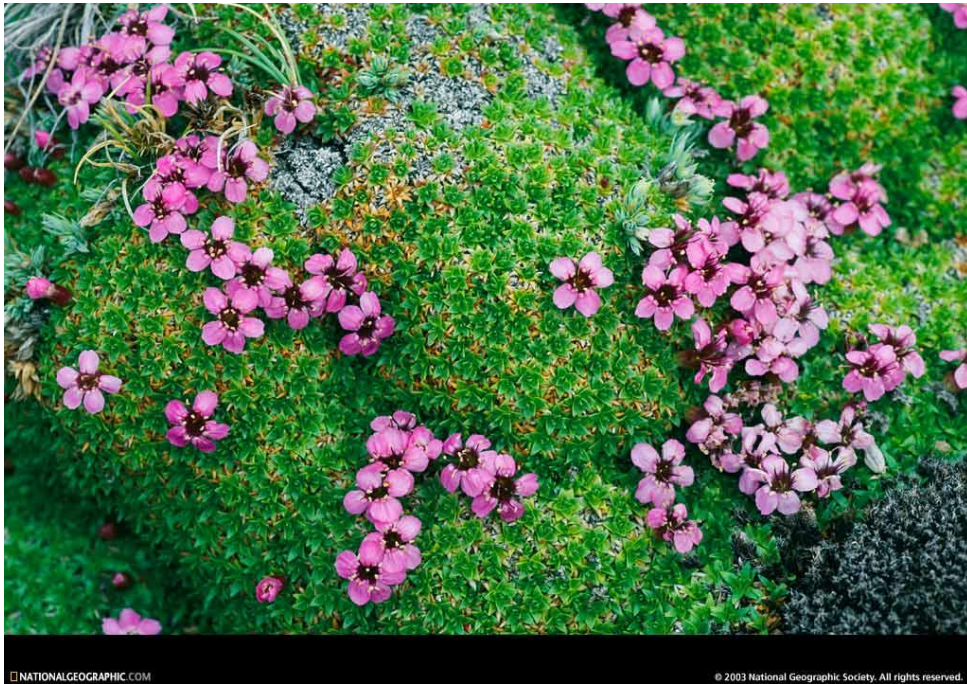


Position and lighting cues: Cast shadows



There are lots of visual cues we can use...

Grouping cues: Similarity (color, texture, proximity)



Grouping cues: “Common fate”



The state of Computer Vision and AI: we are really, really far away.

Oct 22, 2012

<https://karpathy.github.io/2012/10/22/state-of-computer-vision/>

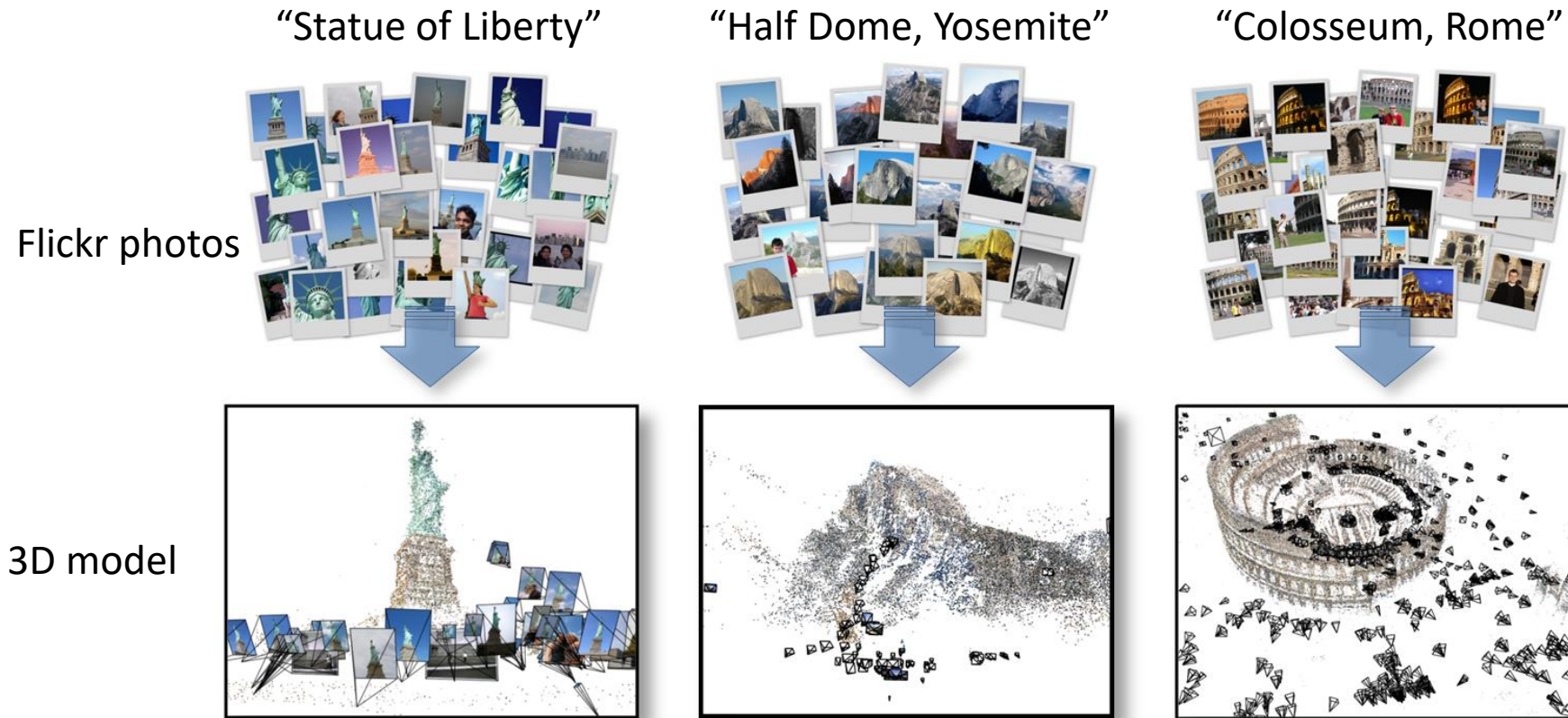


- ❖ Bunch of people in a hallway.
- ❖ There are 3 mirrors in the scene so some of those people are “fake” replicas from different viewpoints.
- ❖ You recognize Obama [...] he is in his suit [...] he is surrounded by other people with suits.
- ❖ There’s a person standing on a scale.
- ❖ Obama has his foot positioned just slightly on top of the scale.
- ❖ Obama is leaning in on the scale, which applies a force on it. Scale measures force that is applied on it => it will over-estimate the weight of the person standing on it.
- ❖ The person measuring his weight is not aware of Obama doing this.
- ❖ People are self-conscious about their weight.
- ❖ There are people in the back who find the person’s imminent confusion funny.
- ❖ The perpetrator is the president, making it even a little more funnier.

Computer Vision Applications

Keep in mind that all these applications can serve as references, guidance, and inspiration when proposing topics for your final course project.

Automatic 3D reconstruction from Internet photo collections



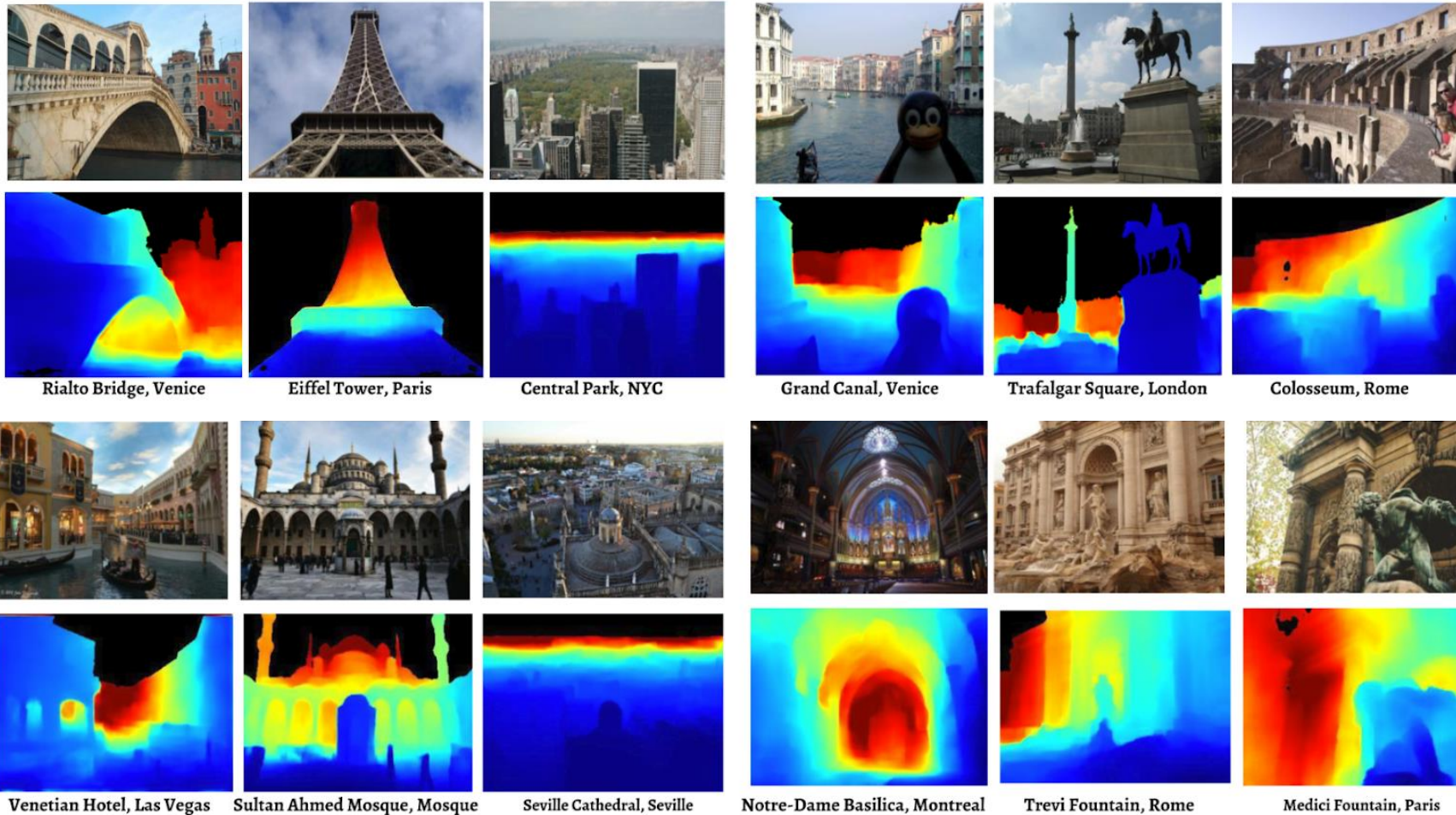
Noah Snavely et al. "Scene reconstruction and visualization from community photo collections."
Proceedings of the IEEE 98.8 (2010): 1370-1390.

City-scale 3D reconstruction (Large-scale Structure-from-Motion)

Reconstruction of Dubrovnik, Croatia, from ~40,000 images (Source: Noah Snavely)

<https://www.youtube.com/watch?v=mUhYJnd1TqE>

Depth estimation from a single image (Monocular Depth Estimation)



Zhengqi Li and Noah Snavely. "MegaDepth: Learning single-view depth prediction from internet photos." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018.

Improve photos ("Computational Photography")



Super-resolution

(W. Yang et al. "Deep learning for single image super-resolution: A brief review." *IEEE Transactions on Multimedia* 21.12 (2019): 3106-3121)



Depth of field on cell phone camera (Picture without (left) and with (right) portrait mode; good way to draw the viewer's attention to a subject)

[Pixel 2](#), [Pixel 3](#) and [Pixel 4](#).



Removing objects

(<https://blog.google/product/s/photos/magic-eraser/>)

Low-light photography

(S. W. Hasinoff et al. "Burst photography for high dynamic range and low-light imaging on mobile cameras." *ACM Transactions on Graphics (ToG)* 35.6 (2016): 1-12)



Slide adapted from
Noah Snavely

Self-driving cars



Source: <https://analyticsindiamag.com/what-is-the-difference-between-computer-vision-and-image-processing/>

Tesla AI Day: <https://www.youtube.com/watch?v=j0z4FweCy4M&t=2814s>

Image generation



“Progressive Growing of GANs for Improved Quality, Stability, and Variation” (Karras et al., 2017)

Which face is real?

Click on the person who is real.



<https://www.whichfaceisreal.com/>

Which face is real?

Click on the person who is real.



<https://www.whichfaceisreal.com/>

Which face is real?

Click on the person who is real.



<https://www.whichfaceisreal.com/>

Image generation (from text prompts)

color photograph of marie curie with a marvel ironman suit



color photograph of batman and queen elizabeth the second of united kingdom



color photograph of a dog in a coral reef



Video generation/animation (from text prompts)



<https://www.youtube.com/watch?v=BTeC4n1veF4>

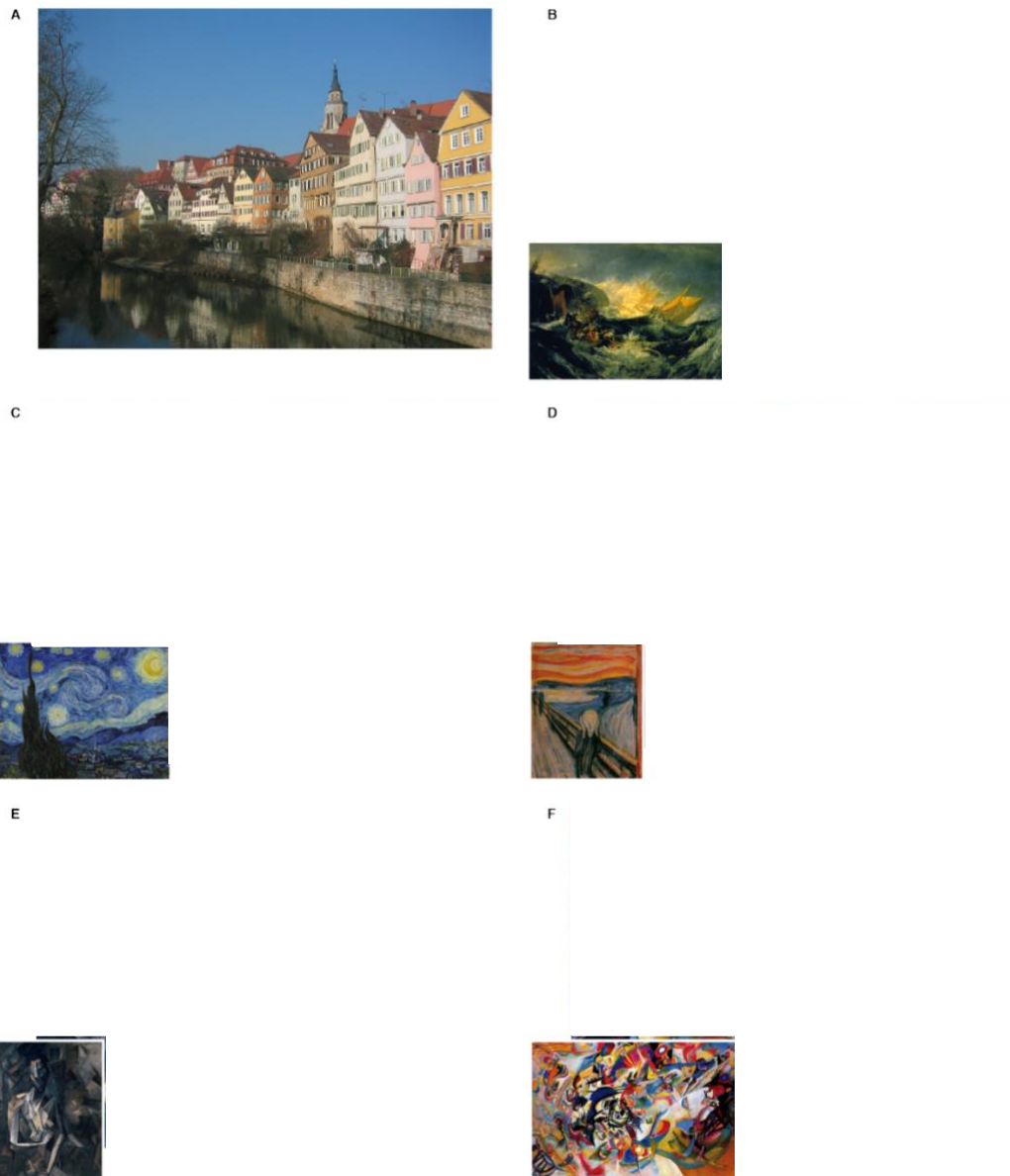
<https://kling.kuaishou.com/en>

Video generation/animation (from text prompts)



Sora: <https://openai.com/index/video-generation-models-as-world-simulators/>

Style Transfer



“A Neural Algorithm of Artistic Style”
(Gatys et al., 2015)

Image colorization



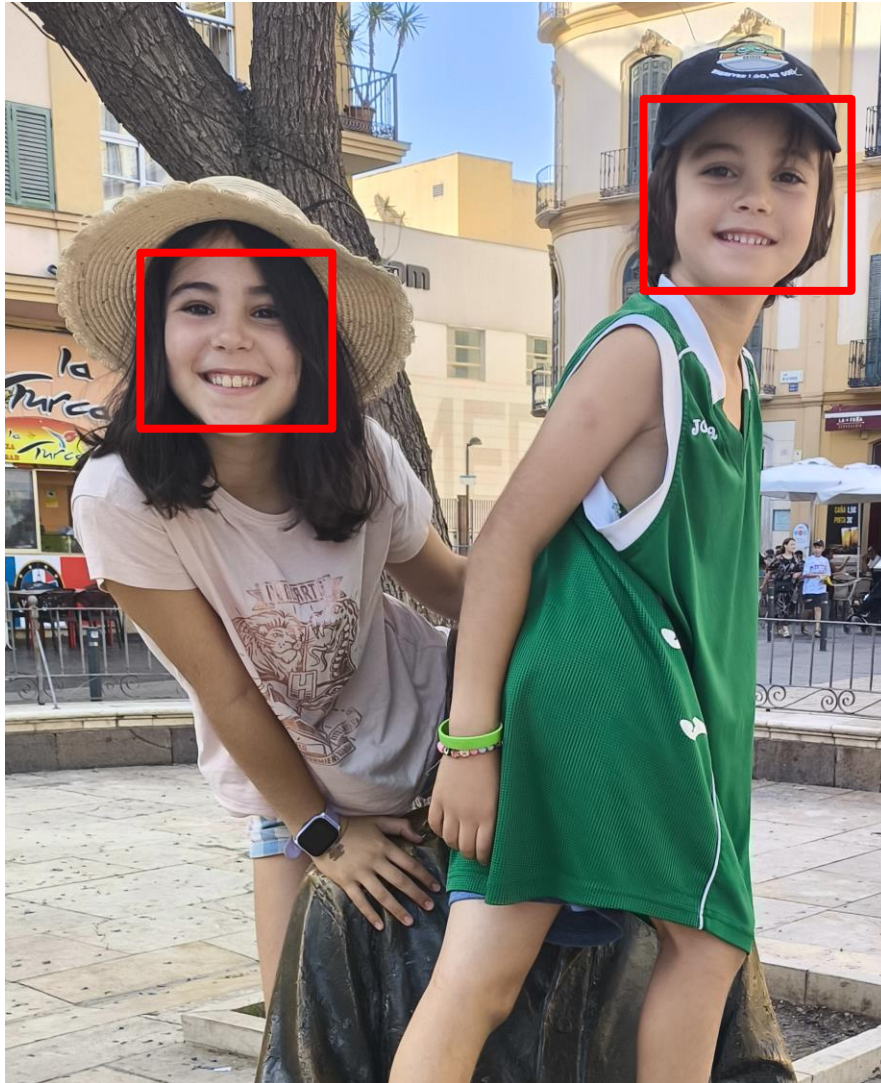
“Colorful Image Colorization” (Zhang et al., 2016)

Image reconstruction/inpainting

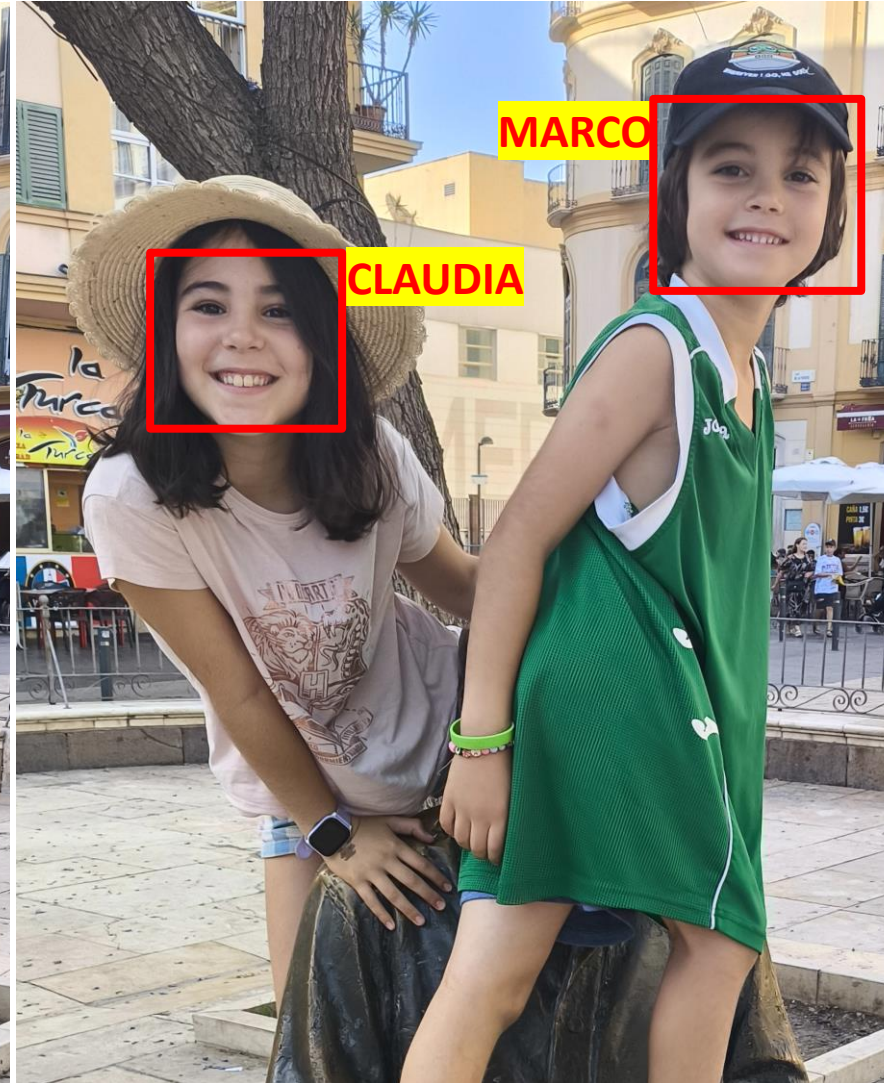


“Image Inpainting for Irregular Holes Using Partial Convolutions” (Liu et al., 2018)

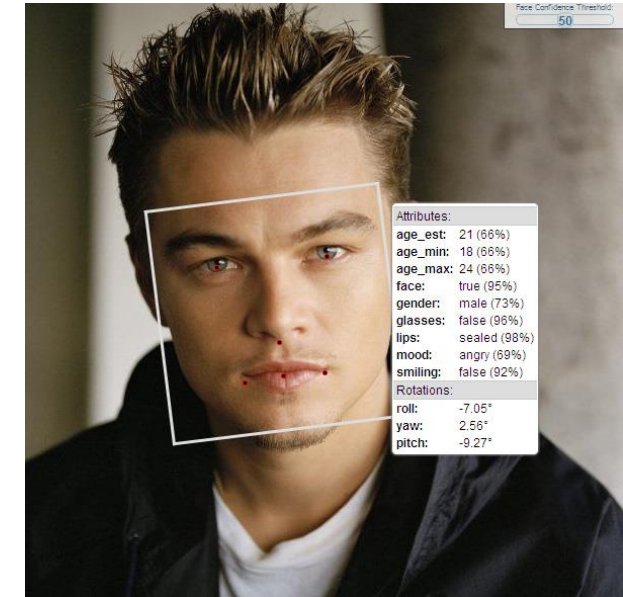
Face detection



Face recognition



Face analysis



Attributes:	
age_est:	21 (66%)
age_min:	18 (66%)
age_max:	24 (66%)
face:	true (95%)
gender:	male (73%)
glasses:	false (96%)
lips:	sealed (98%)
mood:	angry (69%)
smiling:	false (92%)
Rotations:	
roll:	-7.05°
yaw:	2.56°
pitch:	-9.27°

Many companies develop or use face detection, recognition, and analysis technologies.

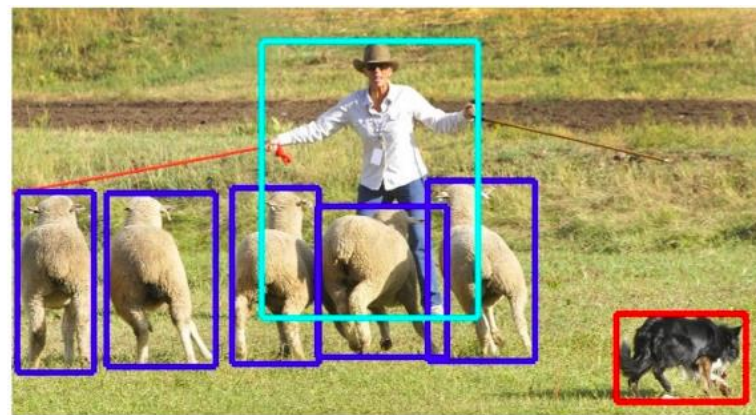
See, for instance,

<https://hertasecurity.com/es/>

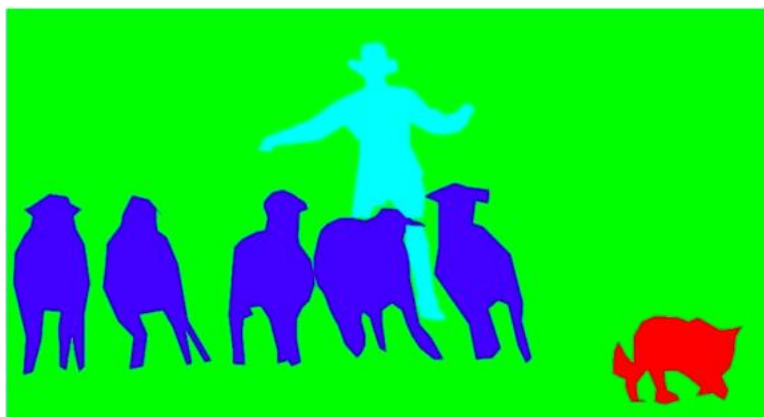
Detection, semantic segmentation, instance segmentation



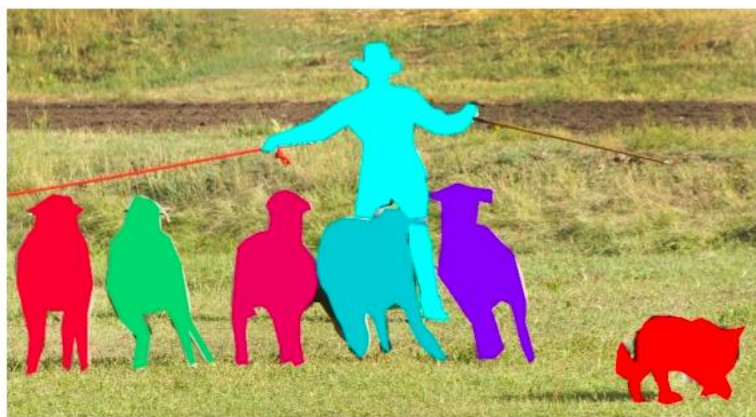
image classification



object detection

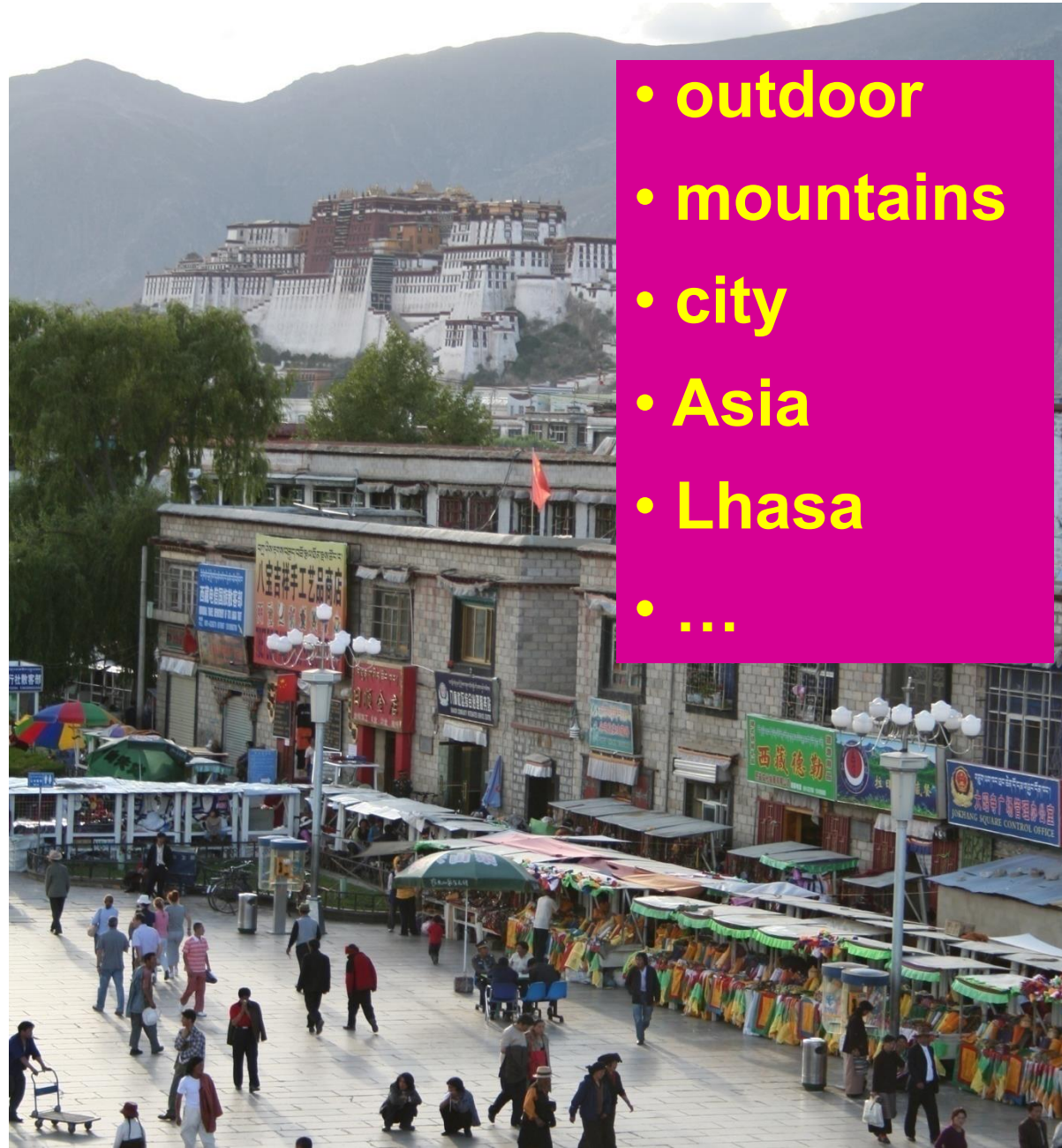


semantic segmentation



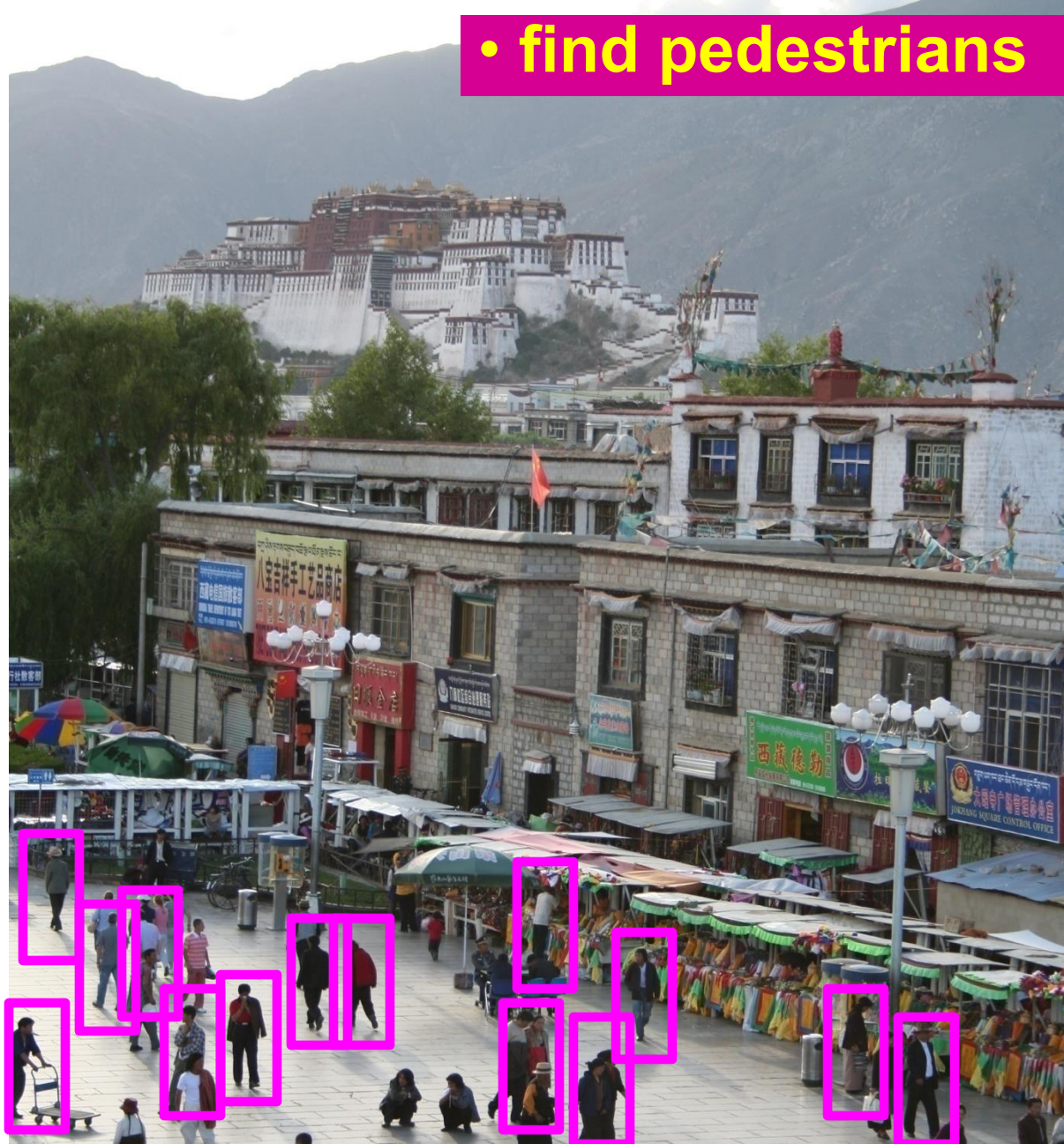
instance segmentation

Image classification and tagging



Object detection

- find pedestrians

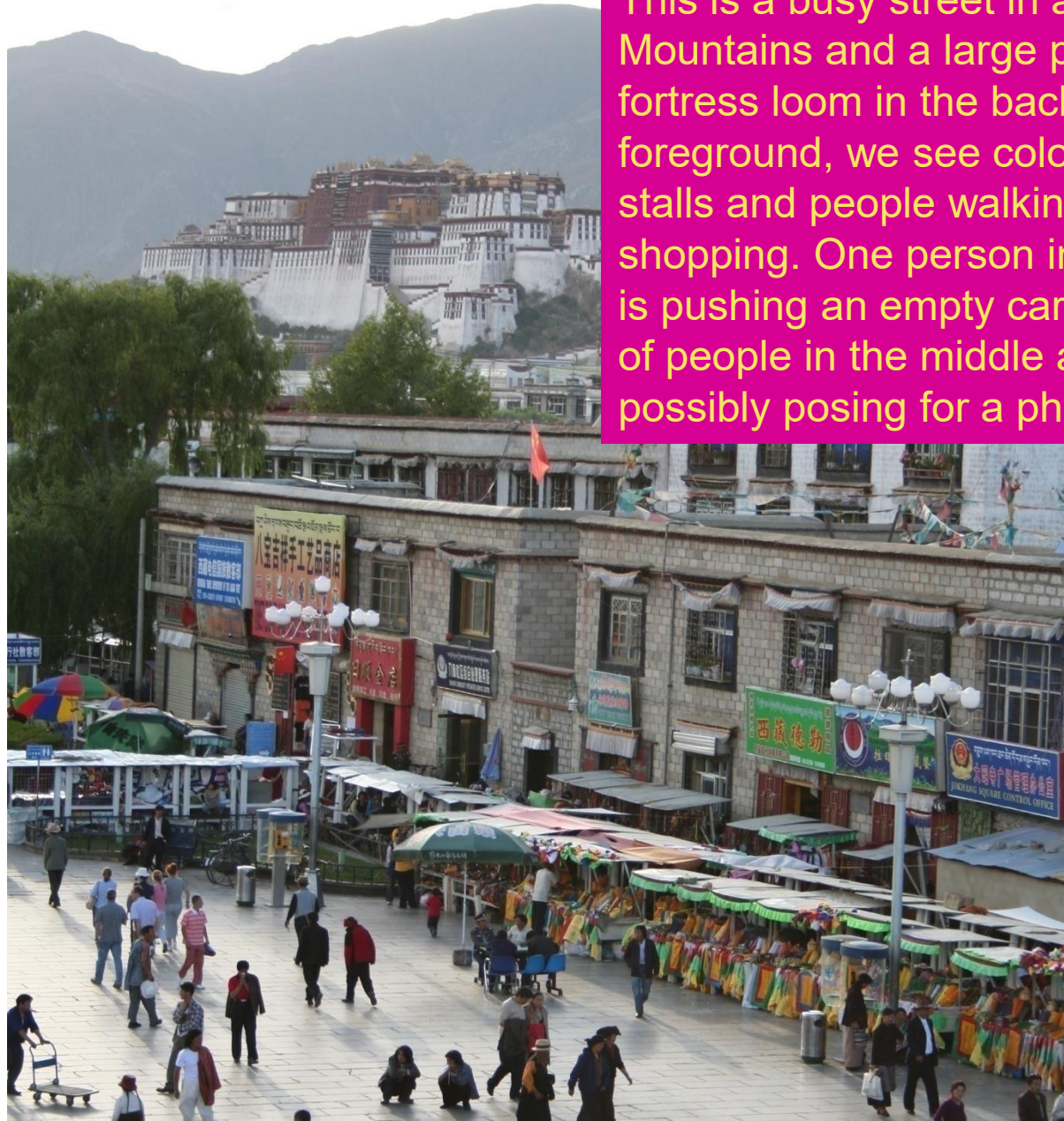


Activity recognition



- walking
- shopping
- rolling a cart
- sitting
- talking
- ...

Image description/captioning



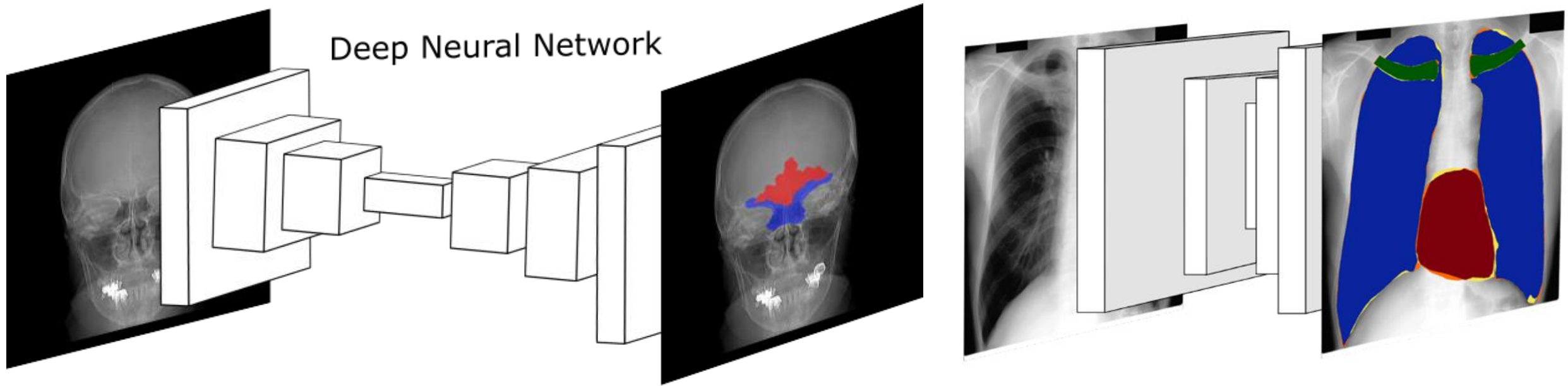
This is a busy street in an Asian city. Mountains and a large palace or fortress loom in the background. In the foreground, we see colorful souvenir stalls and people walking around and shopping. One person in the lower left is pushing an empty cart, and a couple of people in the middle are sitting, possibly posing for a photograph.

Human pose estimation in videos

Real-time Multi-Person 2D Pose Estimation Using Part Affinity Fields

Zhe Cao, Tomas Simon, Shih-En Wei, Yaser Sheikh
Carnegie Mellon University

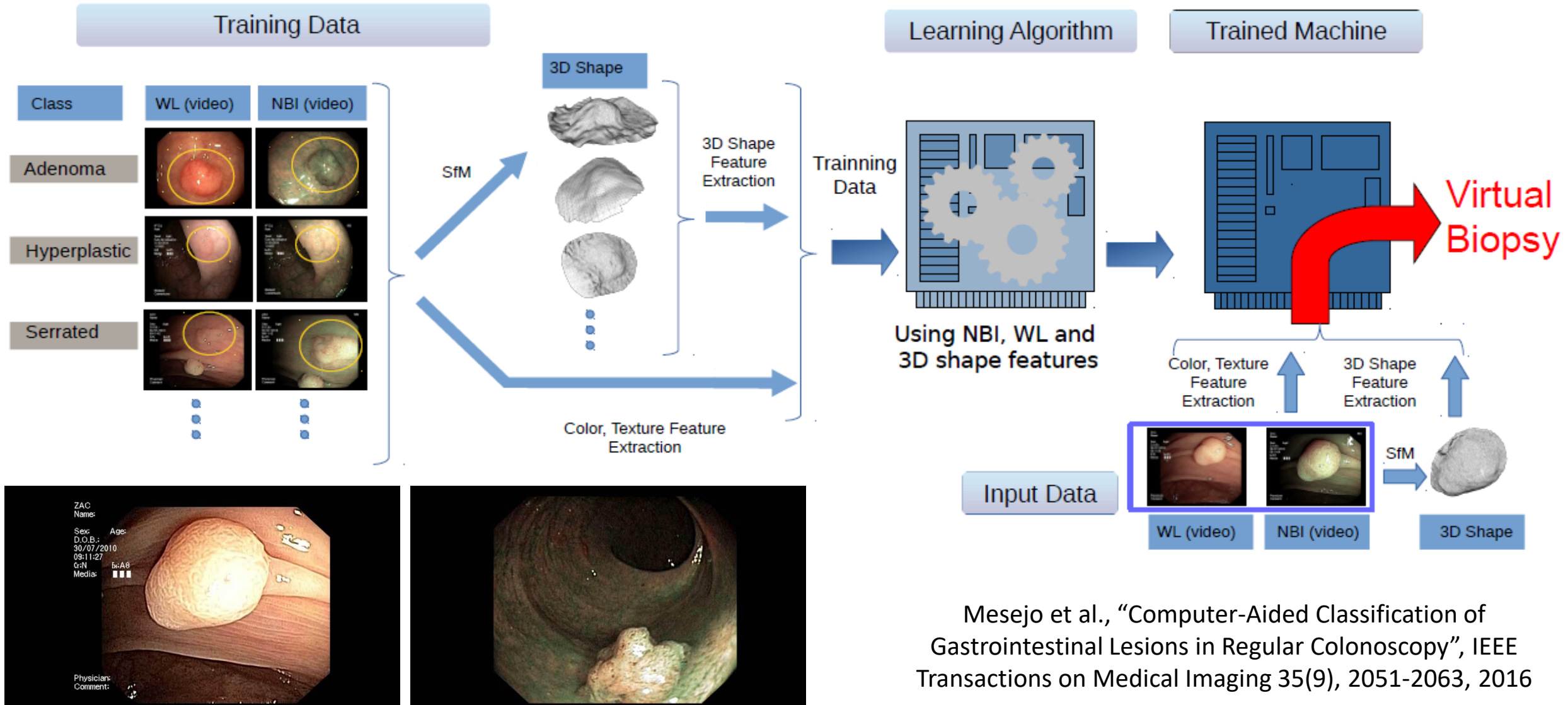
Biomedical Image Segmentation



Gómez et al., “Deep architectures for high-resolution multi-organ chest X-ray image segmentation”, Neural Computing & Applications 32, 15949-15963, Springer, October – 2019

Gómez et al. “Deep architectures for the segmentation of frontal sinuses in X-Ray images: towards an automatic forensic identification system in comparative radiography”, Neurocomputing 456, 575-585, Elsevier, October - 2021

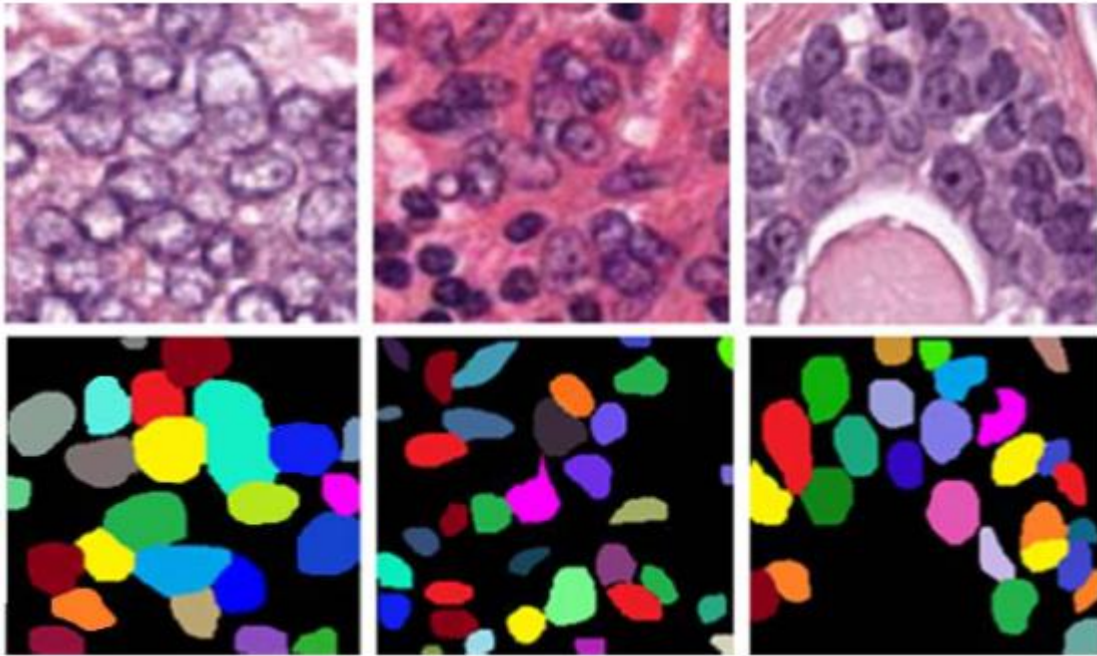
Biomedical Image Classification



Mesejo et al., "Computer-Aided Classification of Gastrointestinal Lesions in Regular Colonoscopy", IEEE Transactions on Medical Imaging 35(9), 2051-2063, 2016

Biomedical Image Analysis

Computational Pathology



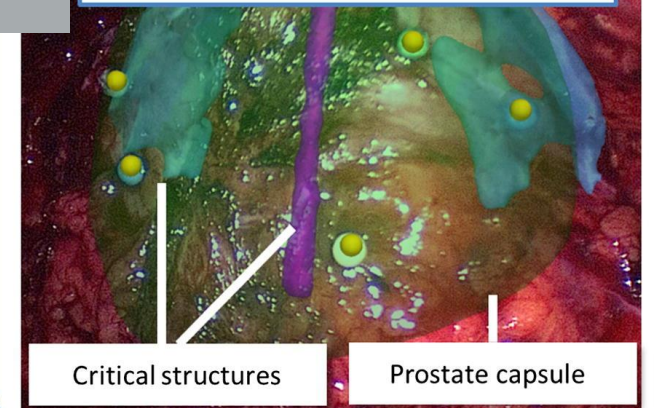
Kumar, Neeraj, et al. "A dataset and a technique for generalized nuclear segmentation for computational pathology." *IEEE transactions on medical imaging* 36.7 (2017): 1550-1560.



Grimson, W. Eric L., et al. "Image-guided surgery." *Scientific American* 280.6 (1999): 62-69.

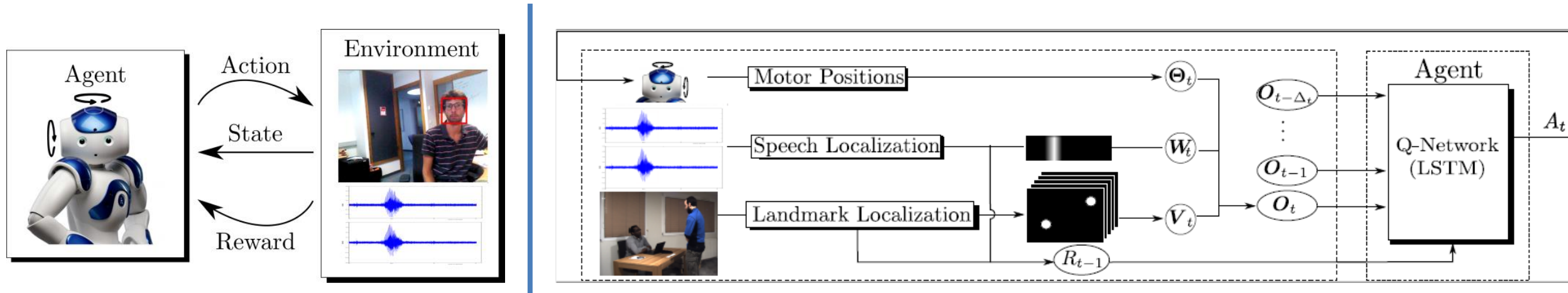
Image-guided Surgery

Augmented Reality Guidance



Maier-Hein, Lena, et al. "Optical techniques for 3D surface reconstruction in computer-assisted laparoscopic surgery." *Medical image analysis* 17.8 (2013): 974-996.

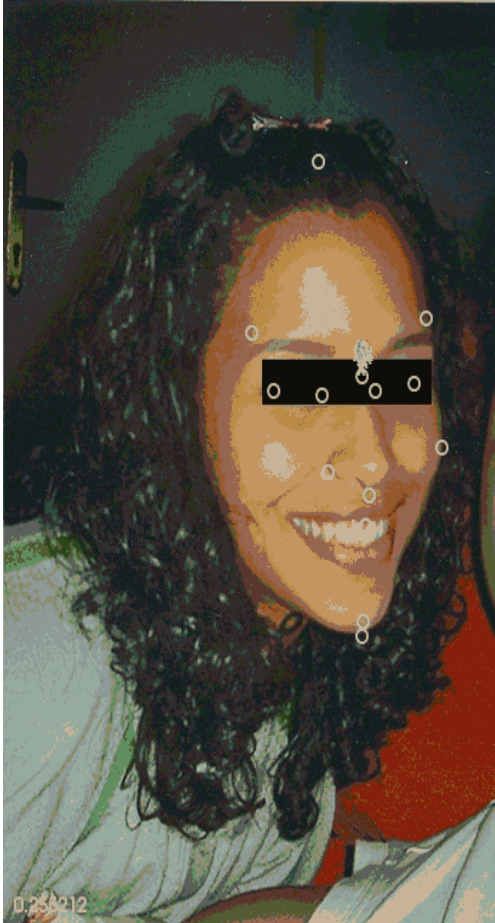
Audio-visual fusion for human-robot interaction



Lathuilière et al., “Neural Network-based Reinforcement Learning for Audio-Visual Gaze Control in Human-Robot Interaction”, Pattern Recognition Letters, 2019

Lathuilière et al., “Deep reinforcement learning for audio-visual gaze control”, IROS, 2018

Forensic Human Identification through Craniofacial Superimposition

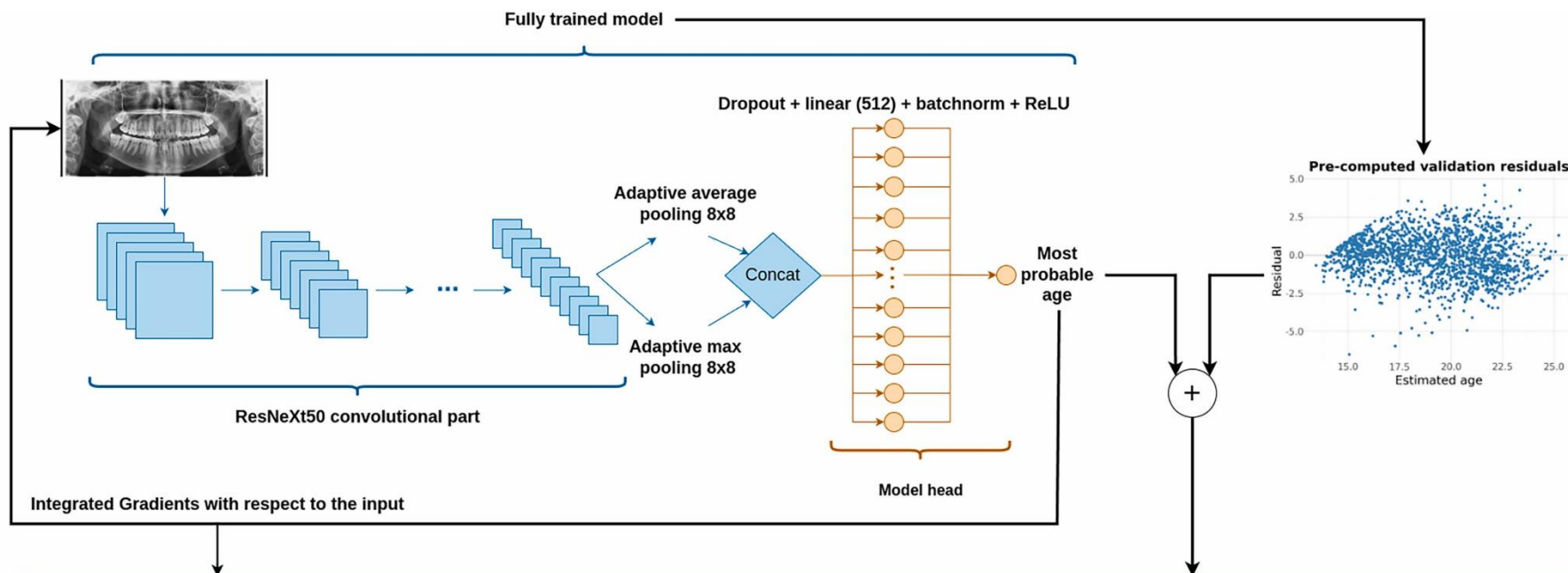


<https://panacea-coop.com>



<https://skeleton-id.com/>

Age Estimation from Dental Panoramic Radiographs



The most probable age for the individual is

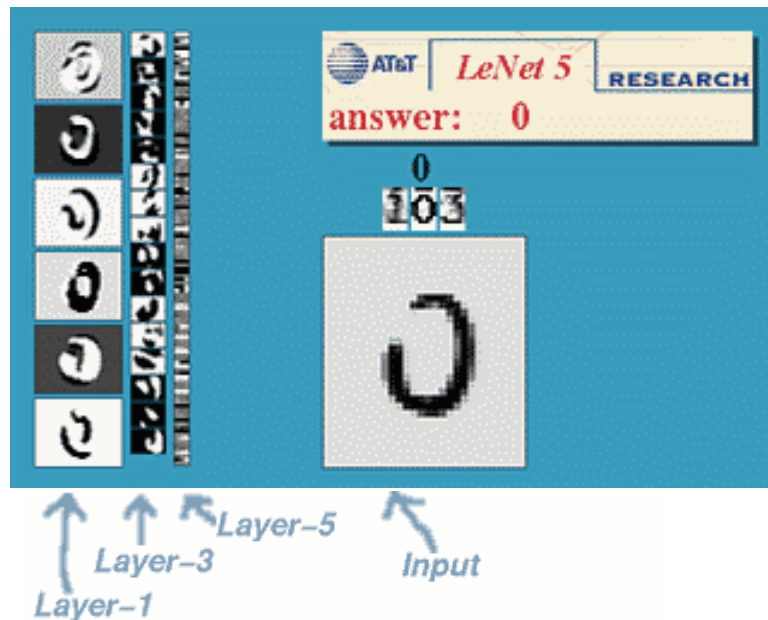
18.24 years

with a minimum age of 15.59 years old according to the 95% prediction interval.

According to the results in our test sample, the individual has a 57% probability of being an adult.

The minimum age of the individual, using prediction intervals for coverage levels of 60, 80, 90, 95 and 99 percent, is 17.14, 16.52, 16.01, 15.59, and 14.57 years old. As a consequence, the maximum coverage for which the predicted age is larger than 18 years old is 0% (this is the coverage employed when the predicted age is below 19.1 years old). In this scenario, the probability of our system correctly classifying the individual as an adult is 88.88%.

Optical Character Recognition (OCR)



Digit recognition, AT&T labs (1980s)

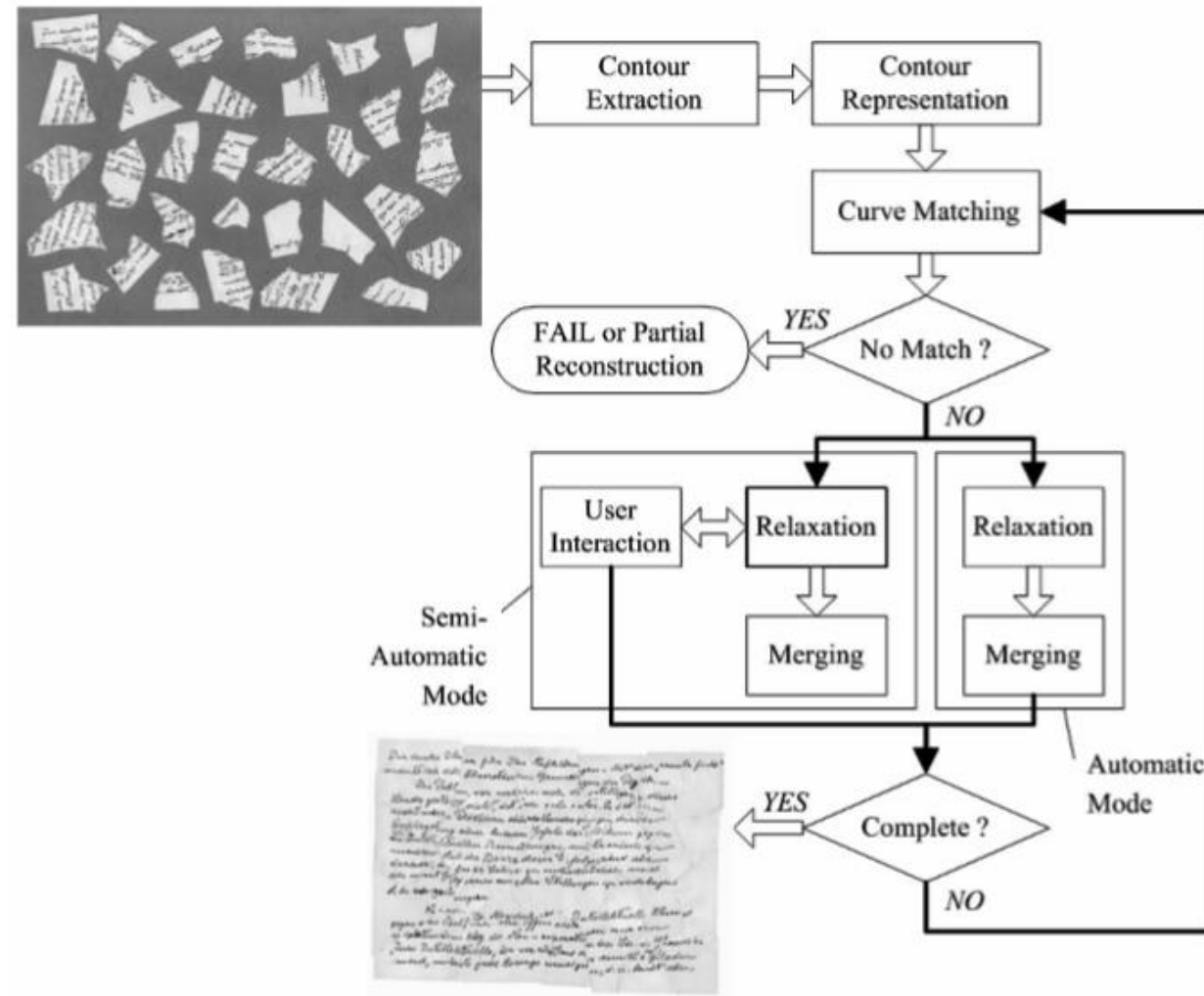
<http://yann.lecun.com/exdb/lenet/>



License plate readers

http://en.wikipedia.org/wiki/Automatic_number_plate_recognition

Reconstruction of Ripped-Up Documents



Zhu, L., Zhou, Z., & Hu, D. (2007). Globally consistent reconstruction of ripped-up documents. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 30(1), 1-13.

Earth viewers (3D modeling)

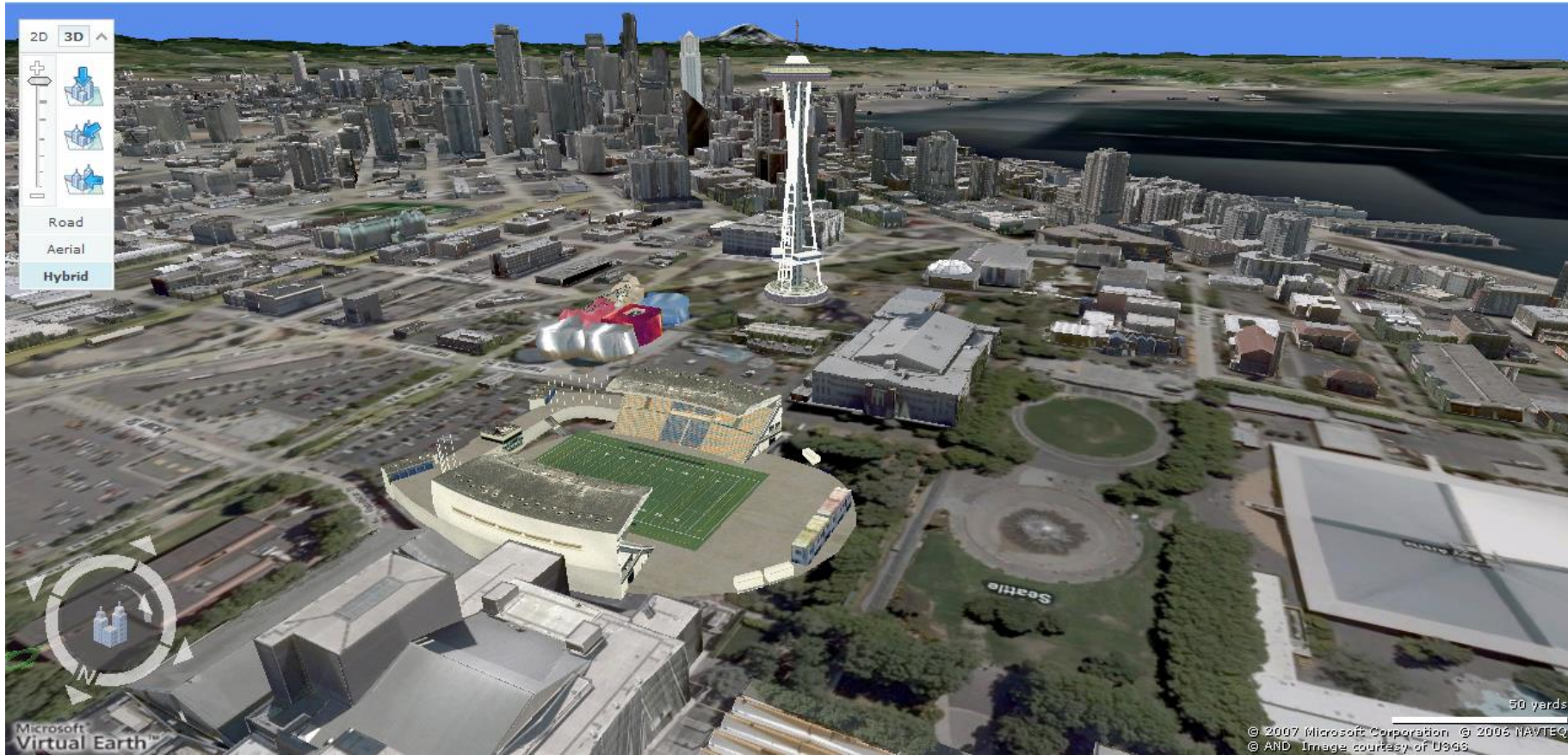


Image from Microsoft's [Virtual Earth](#)
(see also: [Google Earth](#))

Special effects: shape and motion capture



Sports



Sportvision first down line

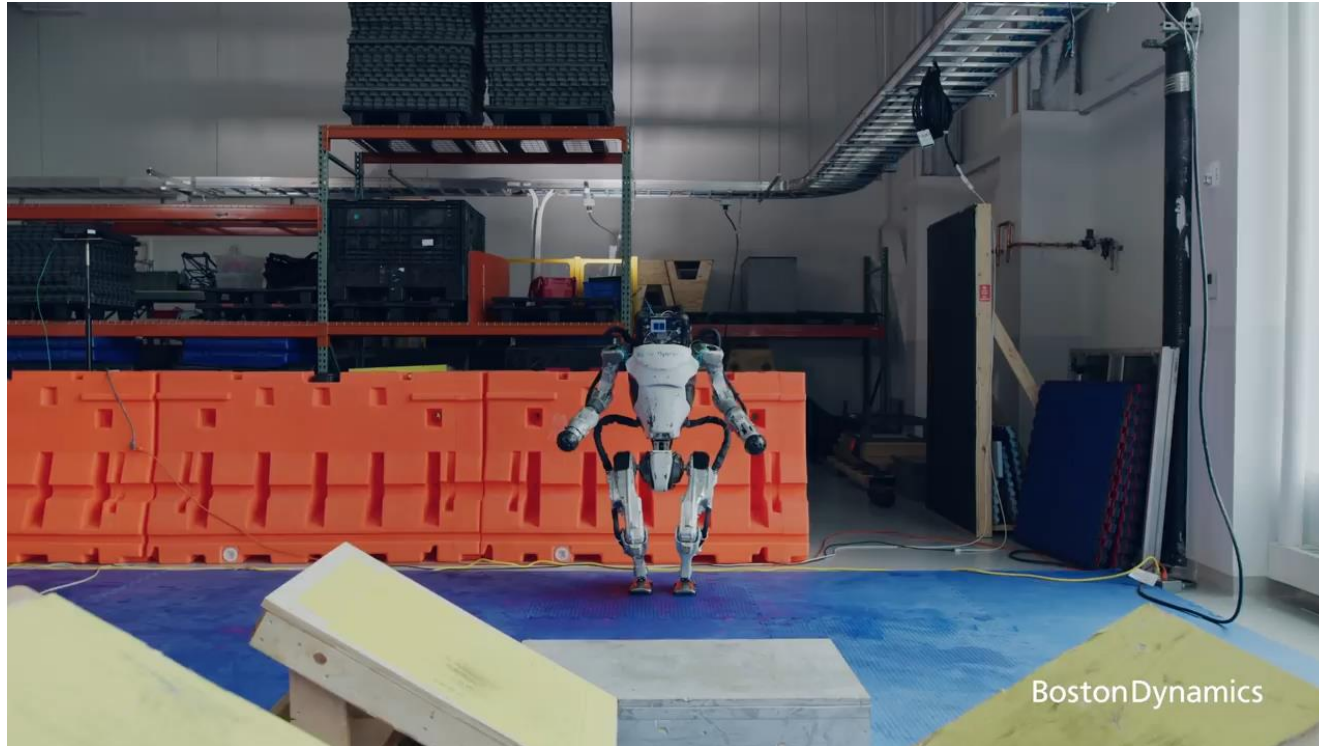
<https://entertainment.howstuffworks.com/first-down-line.htm>



Skillcorner Broadcast Tracking Technology

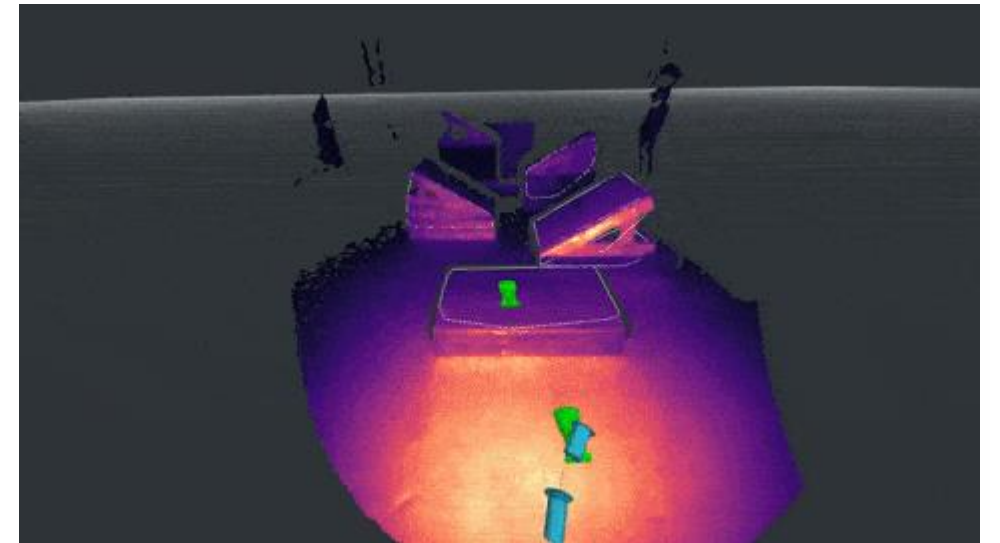
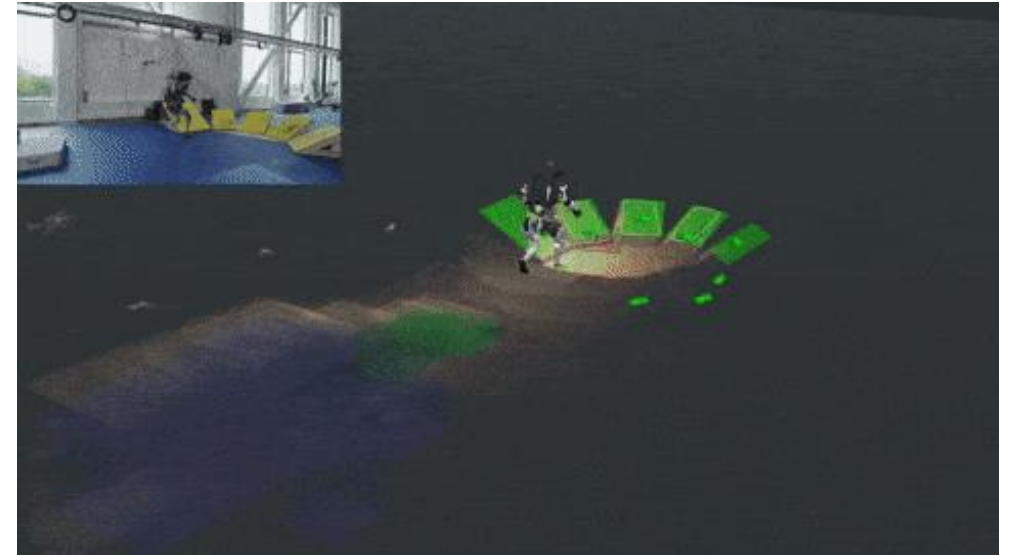
<https://medium.com/skillcorner/a-new-world-of-performance-insight-from-video-tracking-technology-f0d7c0deb767>

Robotics



<https://www.youtube.com/watch?v=tF4DML7FIWk>

<https://bostondynamics.com/blog/flipping-the-script-with-atlas/>



Astronomy



Image created using ChatGPT

Galaxy Classification

Dieleman, S., Willett, K. W., & Dambre, J. (2015). Rotation-invariant convolutional neural networks for galaxy morphology prediction. *Monthly notices of the royal astronomical society*, 450(2), 1441-1459. <https://arxiv.org/pdf/1503.07077>

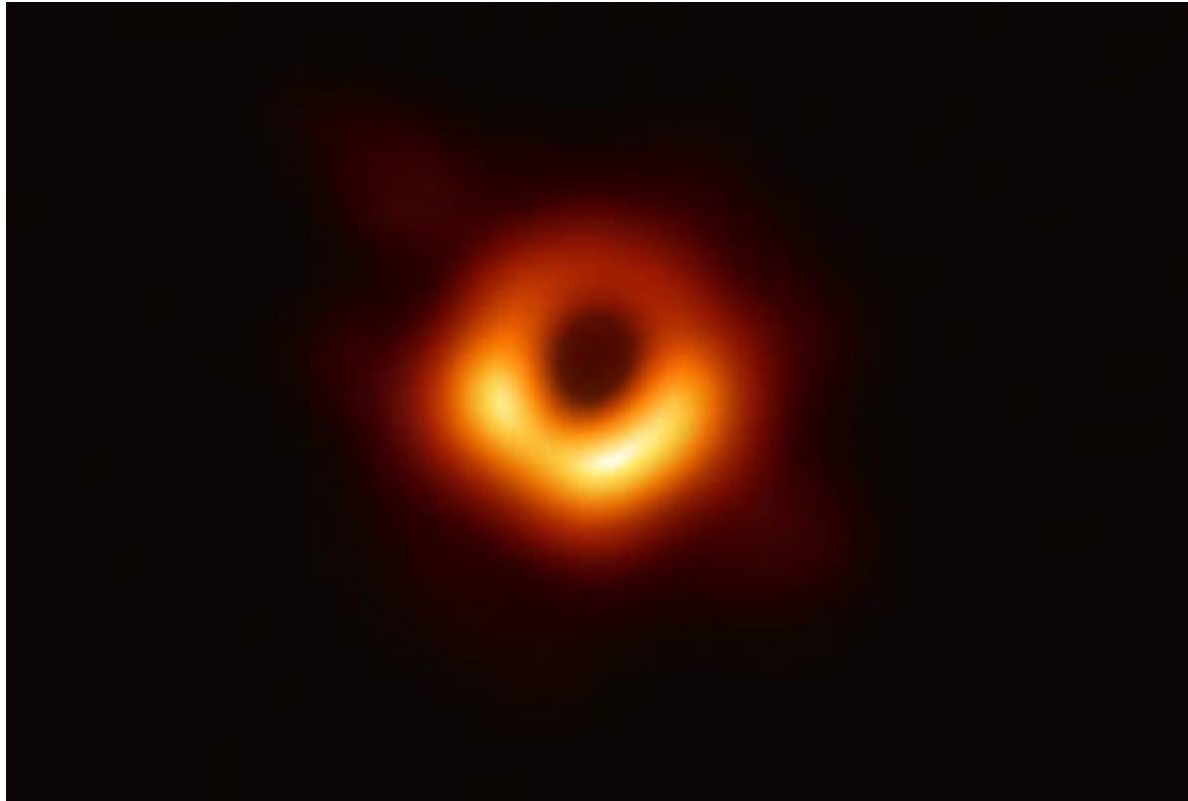
Astronomy

The New York Times

Darkness Visible, Finally: Astronomers Capture First Ever Image of a Black Hole

Astronomers at last have captured a picture of one of the most secretive entities in the cosmos.

April 10, 2019

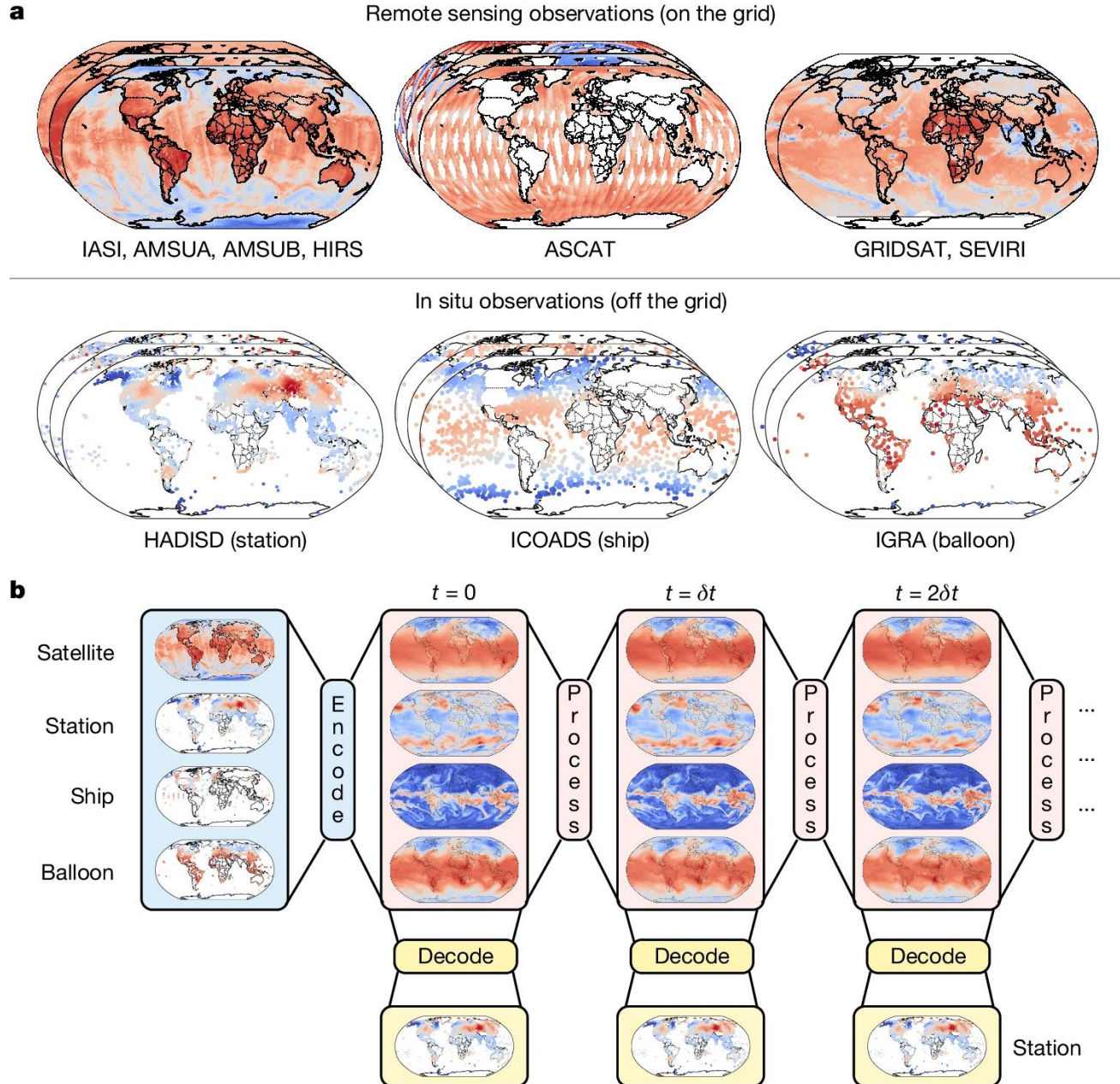


Black Holes | The Edge of All We Know (2020)

<https://www.imdb.com/title/tt11863046/>

Bouman, Katherine L., et al.
"Computational imaging for VLBI image reconstruction." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2016.

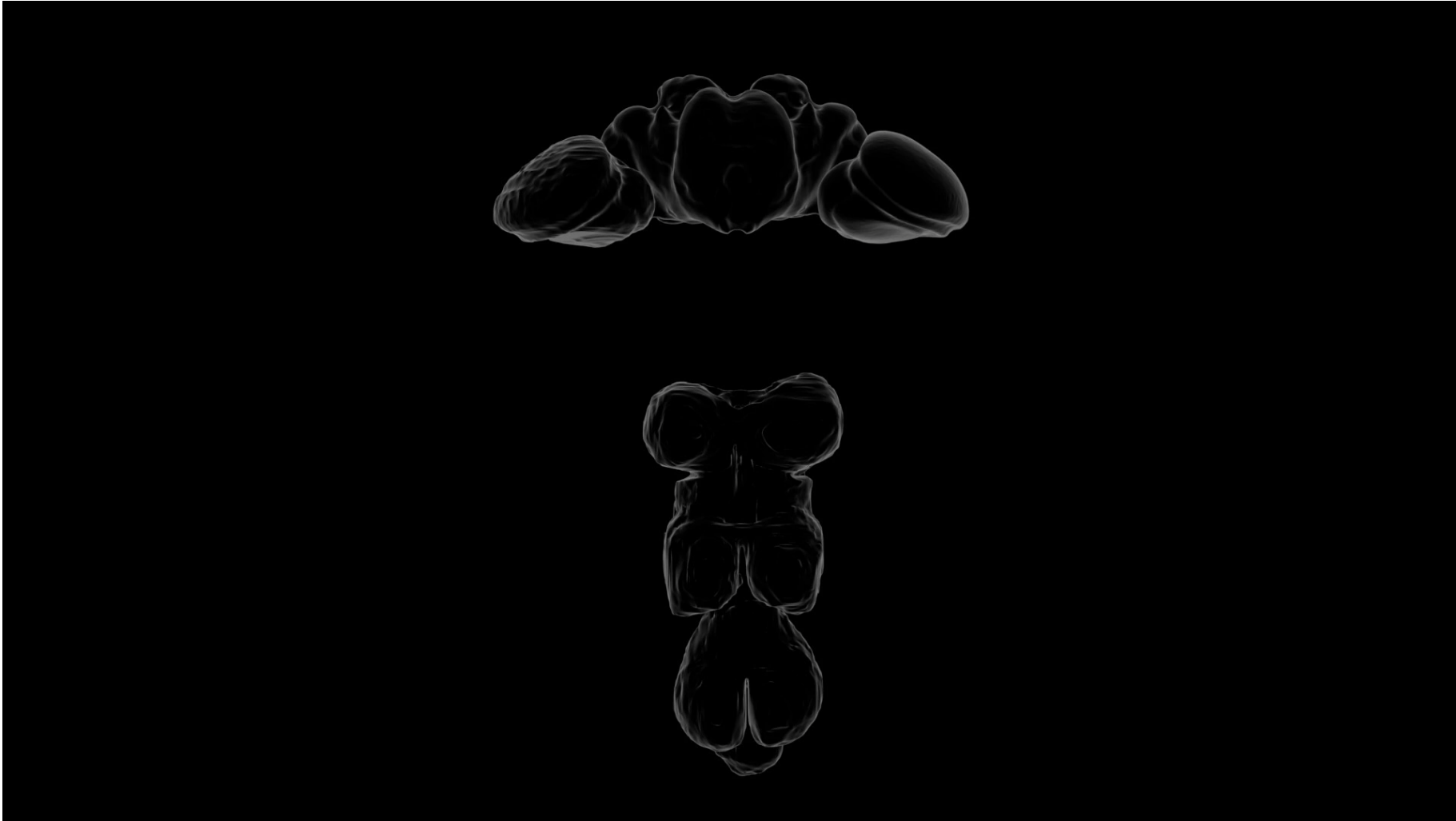
Weather Prediction



“A single machine learning model can replace the entire NWP [numerical weather prediction] pipeline. [...] We believe that Aardvark Weather will be the starting point for a new generation of end-to-end models that will reduce computational costs by orders of magnitude and enable the rapid, affordable creation of customized models for a range of end users.”

Allen, Anna, et al. "End-to-end data-driven weather prediction." *Nature* 641.8065 (2025): 1172-1179.

Neuroscience



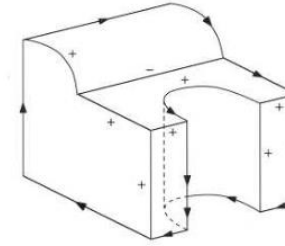
Berg, S., Beckett, I. R., Costa, M., Schlegel, P., Januszewski, M., Marin, E. C., ... & Jefferis, G. S. (2026). Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. *Cell*, 189(18), 5504-5526.

A bit of history

1960s-1970s

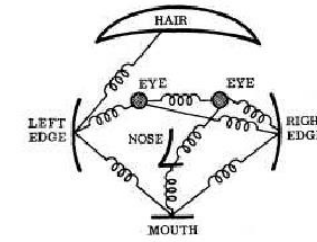
- It was **initially considered an easy step**
 - In 1966, Marvin Minsky at MIT asked his undergraduate student Gerald Jay Sussman to *“spend the summer linking a camera to a computer and getting the computer to describe what it saw”* (Boden 2006, p. 781)
- Recover the **three-dimensional structure** of the world **from images** and to use this as a stepping stone **towards full scene understanding.**

Edge extraction
(line labeling)

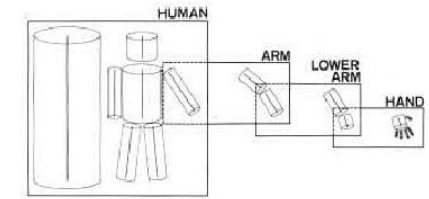


(a)

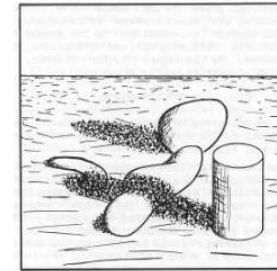
3D modeling of objects



(b)

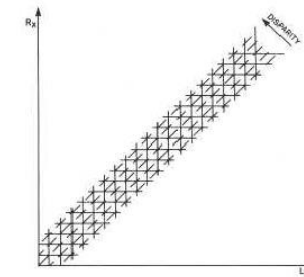


(c)



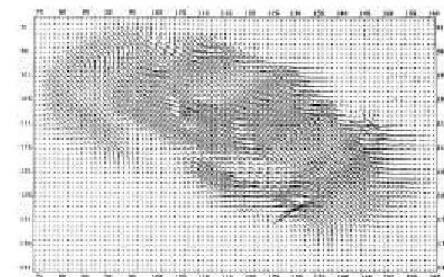
(d)

Understanding
intensities and
shading variations



(e)

Feature-based
stereo
correspondence



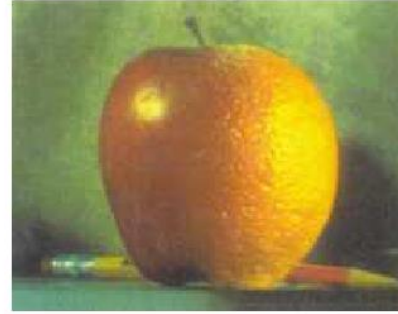
(f)

Intensity-based
optical flow

1980s

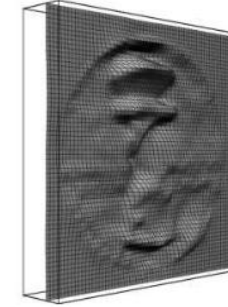
- More sophisticated mathematical techniques for performing **quantitative image and scene analysis**.

Image blending



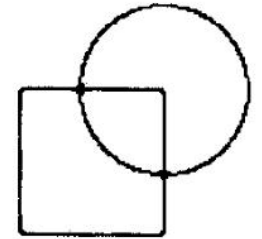
(a)

Shape from shading

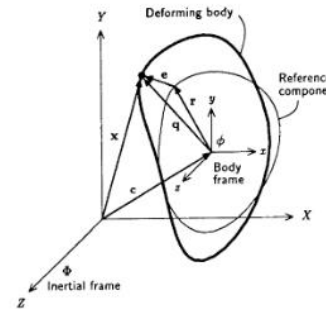


(b)

Edge detection

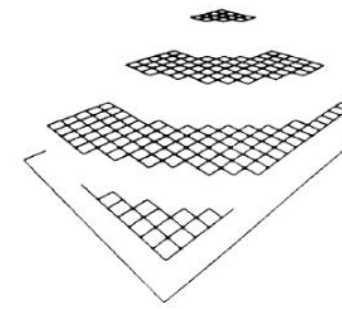


(c)



(d)

Deformable
models



(e)

Regularization-
based surface
reconstruction



(f)

Range data
acquisition and
merging

1990s

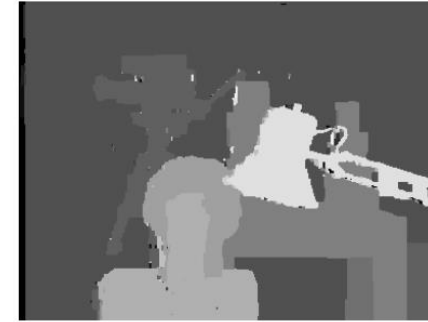
- Much of the previously mentioned topics continued to be explored.
- A lot of the activity was directed at **projective reconstructions**.
- **Multi-view algorithms** that produce **complete 3D surfaces** were also an active topic of research.
- Increased **interaction with computer graphics**.

Structure-from-motion
using orthographic
camera approximations



(a)

Dense stereo
matching

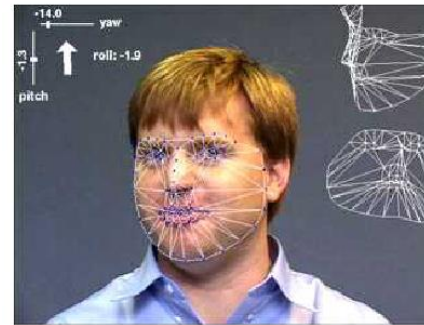


(b)

Multi-view
reconstruction

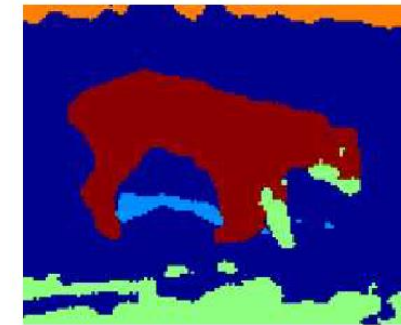


(c)



(d)

Face tracking



(e)

Image
segmentation



(f)

Face recognition

2000s

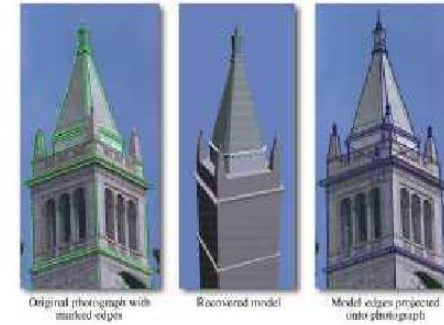
- This decade continued to deepen the **interplay between the vision and graphics fields**, but more importantly embraced **data-driven and learning approaches** as core components of vision.

Image-based rendering (new views of the scene)



(a)

Image-based modeling (generate a 3D model)



(b)

Interactive tone mapping (computational photography)



(c)



(d)

Texture synthesis



(e)

Feature-based techniques for object recognition



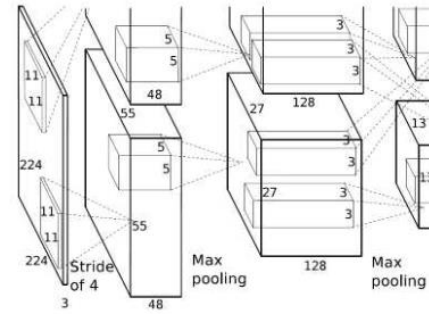
(f)

Region segmentation

- Trend towards using **large labeled (and also self-supervised) datasets** to develop **machine learning** algorithms.
- **Specialized sensors and hardware** for computer vision tasks also continued to advance.
- Incredible advances in the **performance and reliability** of computer vision algorithms.

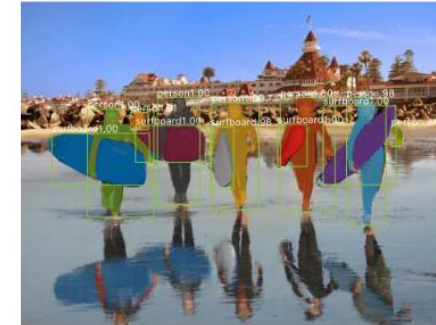
2010s

Deep convolutional neural networks for visual recognition



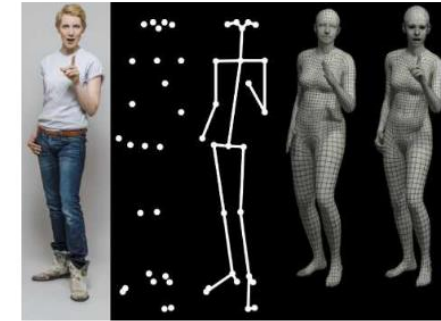
(a)

Object instance segmentation

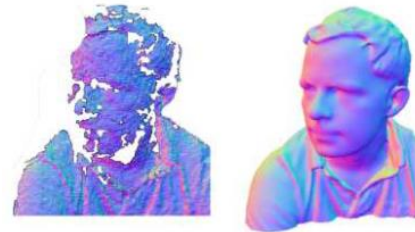


(b)

Person's 3D model with gestures and expression from a single image



(c)



(d)

Fusing multiple color depth images in real-time



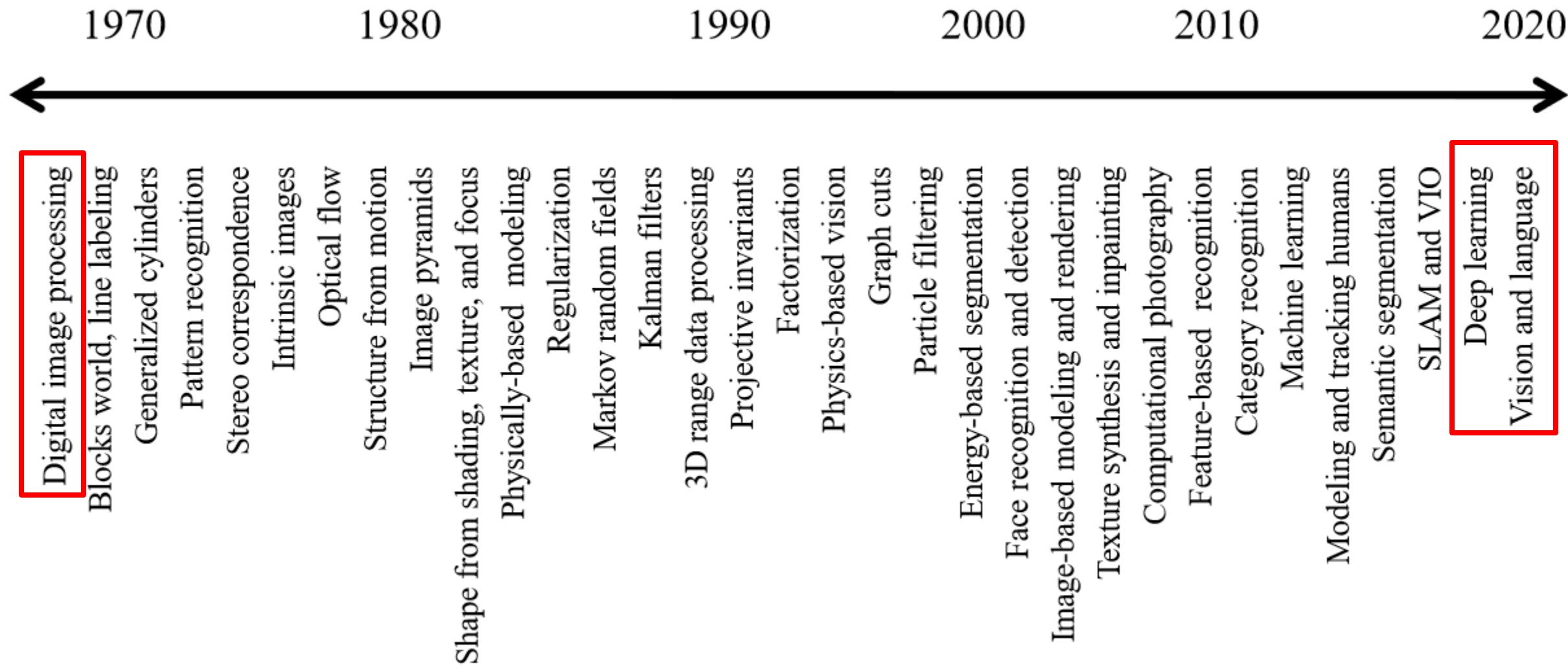
(e)

Smartphone augmented reality



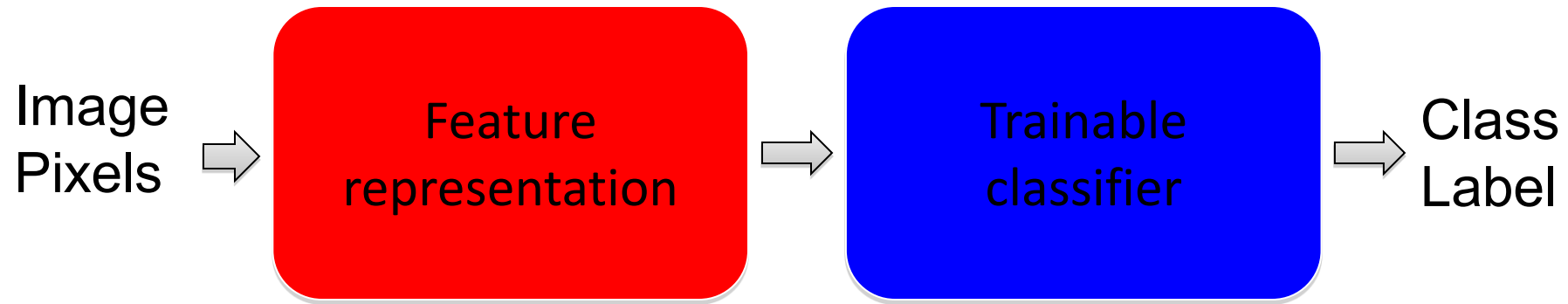
(f)

3D map computed in real-time on a fully autonomous drone



A rough timeline of some of the most active topics of research in computer vision.
(Szeliski, 2022)

“Classic” recognition pipeline



Hand-crafted feature representation

+

Off-the-shelf trainable classifier

Deep Learning



Yoshua Bengio
University of Montreal



Geoffrey Hinton
University of Toronto & Google



Yann LeCun
New York University & Facebook

Turing Award 2018

<https://awards.acm.org/about/2018-turing>

Deep Learning



Fundación
Princesa de Asturias

#PremiosPrincesadeAsturias



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CONCORDIA

SHIGERU BAN



COOPERACIÓN INTERNACIONAL

ELLEN MACARTHUR



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DE REFUGIADOS



INVESTIGACIÓN CIENTÍFICA Y
TÉCNICA

GEOFFREY HINTON, YANN LECUN,
YOSHUA BENGIO Y DEMIS
HASSABIS



LETRAS

JUAN MAYORGA

<https://www.fpa.es/es/especial-2022/premiados/>

Deep Learning

The Nobel Prize in Physics 2024



Ill. Niklas Elmehed © Nobel Prize Outreach

John J. Hopfield

Prize share: 1/2



Ill. Niklas Elmehed © Nobel Prize Outreach

Geoffrey E. Hinton

Prize share: 1/2

The Nobel Prize in Physics 2024 was awarded jointly to John J. Hopfield and Geoffrey E. Hinton "for foundational discoveries and inventions that enable machine learning with artificial neural networks"

The Nobel Prize in Chemistry 2024



Ill. Niklas Elmehed © Nobel Prize Outreach

David Baker

Prize share: 1/2



Ill. Niklas Elmehed © Nobel Prize Outreach

Demis Hassabis

Prize share: 1/4



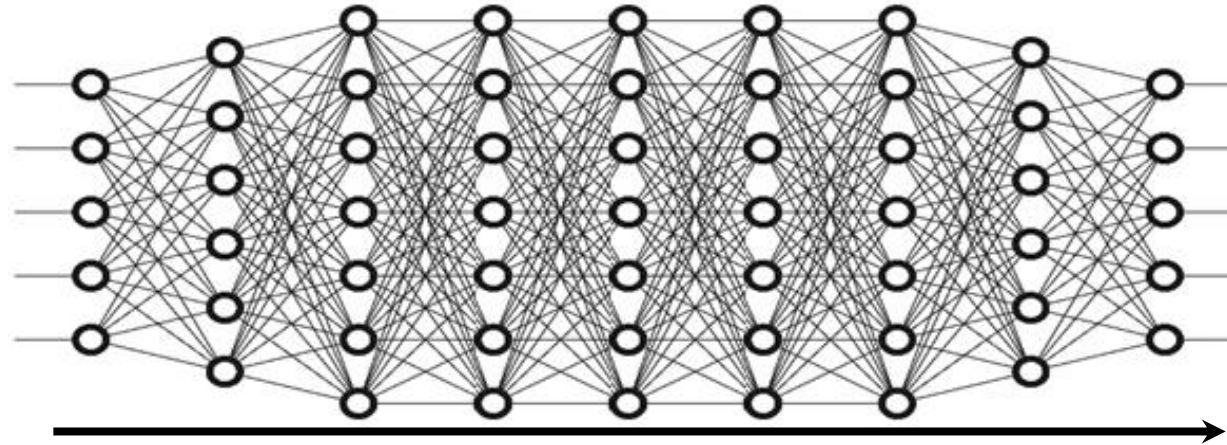
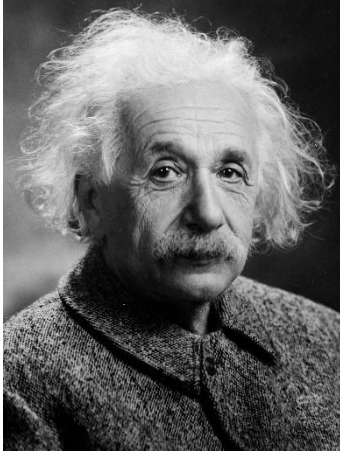
Ill. Niklas Elmehed © Nobel Prize Outreach

John M. Jumper

Prize share: 1/4

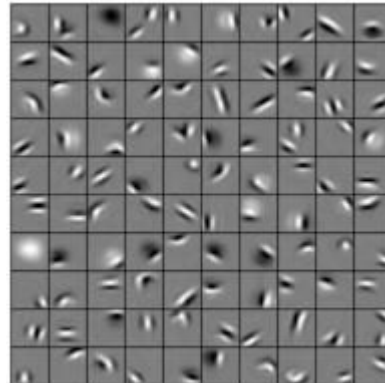
The Nobel Prize in Chemistry 2024 was divided, one half awarded to David Baker "for computational protein design", the other half jointly to Demis Hassabis and John M. Jumper "for protein structure prediction"

Hierarchical Learning

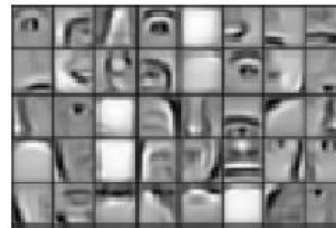


“Albert Einstein”

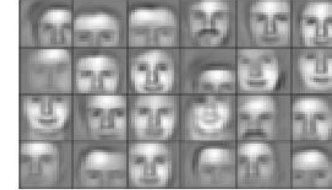
Low-level Features



Mid-level Features



High-level Features



Images taken from Lee et al.
["Convolutional deep belief networks for scalable unsupervised learning of hierarchical representations." ICML 2009.](#)

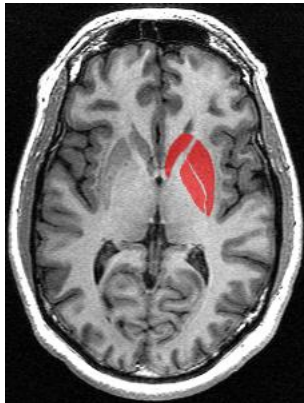
Deep learning allows computational models that are composed of **multiple processing layers to learn representations of data with multiple levels of abstraction.**

Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. "Deep learning." *Nature* 521.7553 (2015): 436-444.

Some of the Main Current Limitations

Some of the Main Current Limitations

- Machines can outperform humans at **concrete, well-defined tasks**, but they struggle at more general and abstract tasks.



Some of the Main Current Limitations (2)

- Many times, your system is hard to analyze (**black box**) and it's only going to be **as good as the data you have**.
 - Serious ethical, practical, legal problems...

Tank



...



No tank



...



Tank?



Recommended reading:
"The Neural Net Tank Urban Legend" <https://gvern.net/tank>

Some of the Main Current Limitations (3)

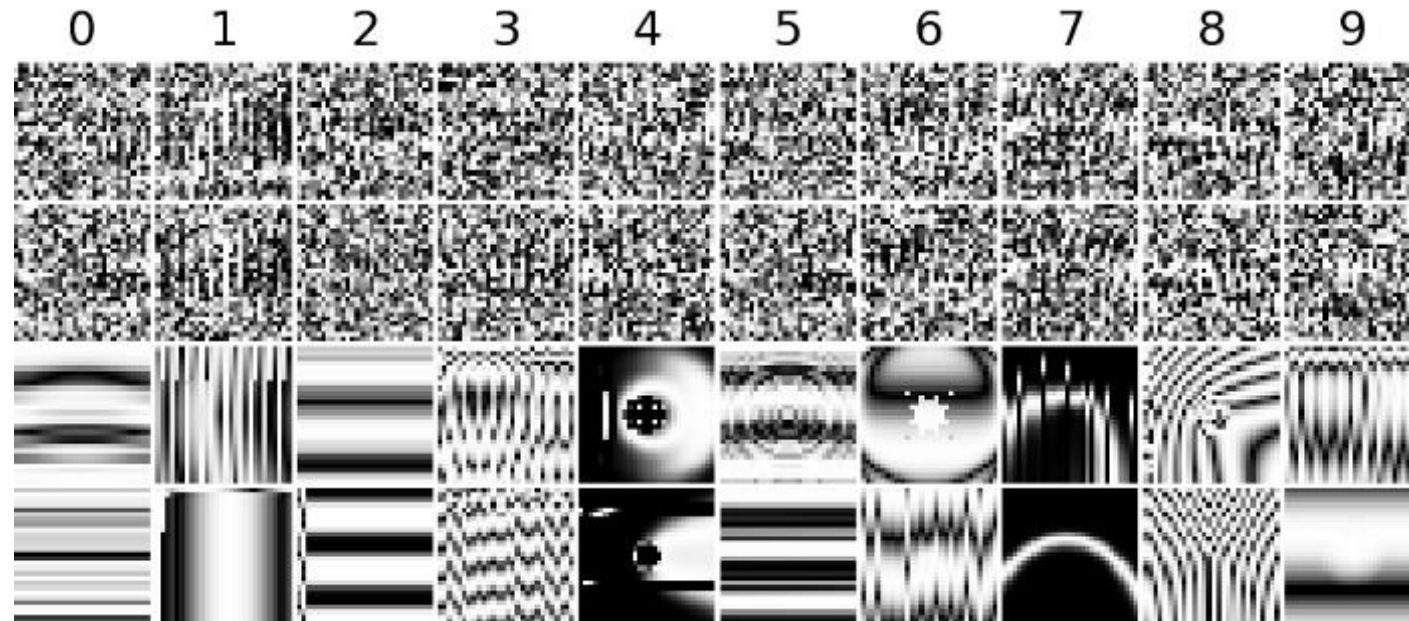
- **Deep learning generally requires too much data (and energy)**
 - Humans don't usually need to see 5000 images of different cats to reliably identify one.



Image taken from <https://www.fourpaws.com/>

Some of the Main Current Limitations (and 4)

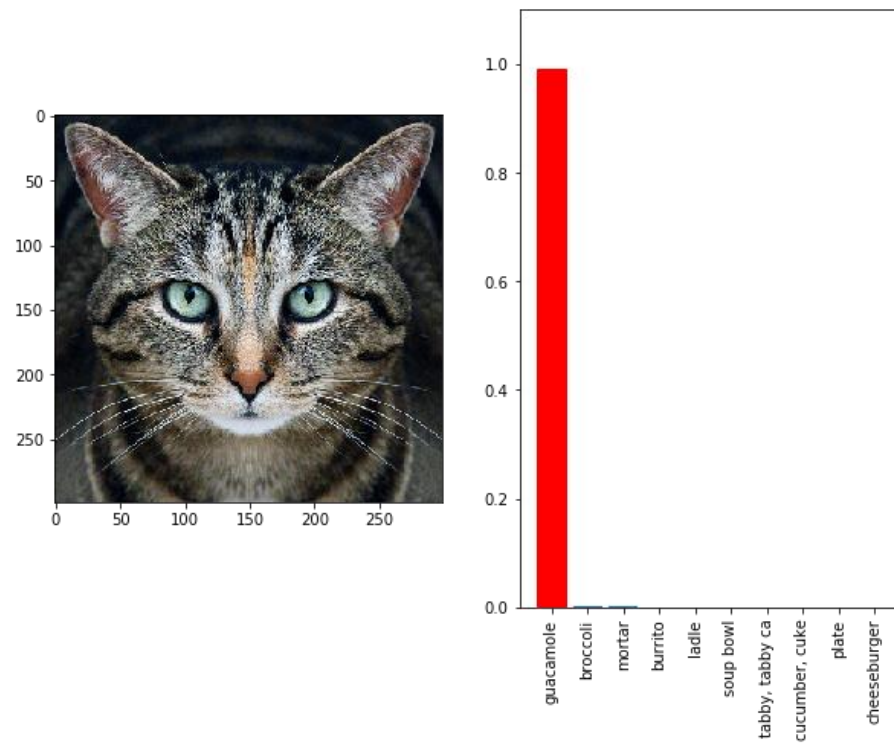
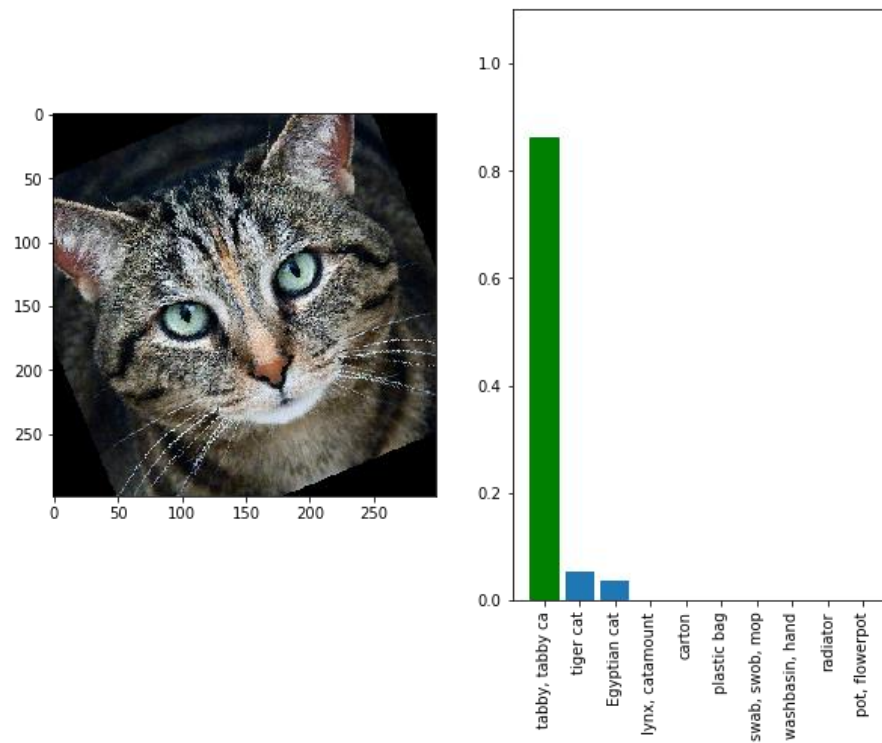
- **Machines have limited semantic grounding**
 - They rely on learned statistical patterns
 - They can struggle to “see the whole”



Deep neural networks are easily fooled: High confidence predictions for unrecognizable images (CVPR 2015)

Some of the Main Current Limitations (and 4)

- **Machines have limited semantic grounding**
 - They rely on learned statistical patterns
 - They can struggle to “see the whole”



Some of the Main Current Limitations (and 4)

- **Machines have limited semantic grounding**
 - They rely on learned statistical patterns
 - They can struggle to “see the whole”



For more recent work on the current challenges regarding adversarial robustness in traffic sign recognition, see:

- [Zhou et al.](#), IEEE TIFS, 2024
- [Khan et al.](#), CVPRW, 2025
- [Fawole & Rawat](#), ACM CSUR, 2025
- [Andleeb et al.](#), IJCCE, 2026

Some of the Main Current Limitations (and 4)

- **Machines lack a physical model of the world**



Prompt: Archeologists discover a generic plastic chair in the desert, excavating and dusting it with great care.

Weakness: In this example, Sora fails to model the chair as a rigid object, leading to **inaccurate physical interactions**.



Prompt: Five gray wolf pups frolicking and chasing each other around a remote gravel road, surrounded by grass. The pups run and leap, chasing each other, and nipping at each other, playing.

Weakness: Animals or people can **spontaneously appear**, especially in **scenes containing many entities**.

<https://openai.com/sora> (15 Feb 2024)
<https://openai.com/research/video-generation-models-as-world-simulators>

Take-home message

Despite the obvious successes achieved, current AI systems:

- Often require **large amounts of data** to learn about the world
- Can require **substantial computational resources and energy**
- Can **fail unexpectedly** or be **fooled by surprisingly simple perturbations**
- Perform best on **well-defined tasks**
- Their **behavior can be difficult to analyze** and understand

“We just don’t have machines with common sense”

(Yann LeCun, 2019)

**Machines, no matter how sophisticated they are, do not solve problems,
they execute solutions**

Introduction

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Departamento de Ciencias de la Computación e Inteligencia Artificial



UNIVERSIDAD
DE GRANADA



DaSCI

Instituto Andaluz de Investigación en
Data Science and Computational Intelligence